

One battle after another

Revamping the U.S. defense industrial base

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EXPLORE

- The U.S. remains the world's apex military power, but its defense industrial base (DIB) is concentrated in a small number of large firms focused on a narrow set of flagship platforms.
- Since the end of the Cold War, amid uncertain defense budgets, the U.S. has prioritized high-performance systems like the F-35—highly capable of projecting power, but slow to develop, costly, and difficult to scale.
- Over the past two decades, the threat landscape has diversified, spanning near-peer competitors, mid-tier states armed with low-cost precision weapons, and non-state actors leveraging commercial technologies.
- Wars in Ukraine and the Middle East, as well as China's Anti-Access/Area-Denial (A2AD) strategy, have amplified these dynamics, reinforcing the need for adaptability.
- Technological superiority alone is no longer sufficient; military advantage increasingly hinges on industrial capacity, cost discipline, and the speed and adaptability of capability deployment.
- The Iran and Ukraine wars also highlight the operational value of low-cost, expendable systems such as drones, alongside command-and-control architectures that can iterate rapidly based on real-time feedback.
- Modern warfare is increasingly defined by the ability to “sense, make sense, and act,” depending on processing vast amounts of data and translating it into timely decisions.
- Speed and adaptability are now critical differentiators, particularly in battlefield environments shaped by cyber warfare and A2AD tactics.
- While hardware remains the foundation of military power, software has emerged as the key enabler, integrating systems, accelerating decision-making, and allowing continuous upgrades.
- This shift demands a new industrial model: advanced but adaptable platforms, scalable and expendable hardware, and software that is continuously updated.
- The DIB needs to fuse with the commercial sector, leveraging its capacity, technology, and the speed of innovation needed for modern warfare, which are also arguably the U.S.'s most durable advantages.
- A new ecosystem of unmanned, attritable, and software-defined systems is rising alongside the U.S.'s dominant exquisite tier, but the two need to be integrated into one rather than left to evolve on parallel tracks.
- This productive shift should be paired with institutional reform, and while commercial-first and software-centric policies are gaining momentum, execution remains a heavy constraint; without follow-through, the U.S. risks its apex position in military power.

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Contents

Arsenal at risk	2
The U.S. defense industrial base	6
Lessons from Other DIBs	10
Asymmetric warfare	23
The Imperatives: Integration, Scale, and Innovation	24
New names challenge incumbent primes	28
Institutional Challenges	31
Future-proofing U.S. defense	32
Appendix	34
I. U.S. and Chinese Military Capabilities	34
II. Drones & Robotics in Modern Warfare	37
III. The F-35 - A Signature Program	41
IV. “High and Low” Strategy - The Barracuda Missiles	44
V. Main global public defense companies*	46
VI. Main U.S. private defense companies	51

See page 53 for important disclosures.

Arsenal at risk

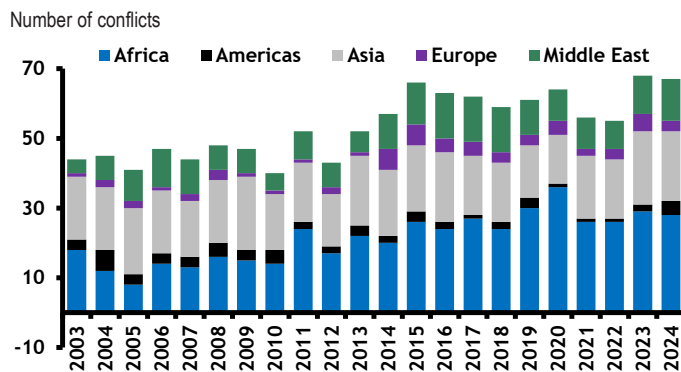
In the early 1990s, it seemed the world had turned a corner. Francis Fukuyama's "[End of History](#)" captured the post-Cold War optimism that, with the collapse of the Soviet Union and the rapid spread of globalization, liberal democracy and market capitalism had emerged as the clear victors in the ideological struggle, making large-scale conflicts between major powers seem obsolete.

But history did not end; it just got messier. In the years that followed, the United States no longer confronted a single, easily defined adversary. Instead, it has had to contend with a diverse array of rivals, from advanced militaries equipped with large arsenals, to revisionist states prepared to endure hardship and escalate tensions, to regional players and proxies skilled in causing disruption. Today's competition spans everything from gray-zone tactics and cyber and electronic warfare to precision strikes, often all at the same time.

Rising conflicts; rising costs

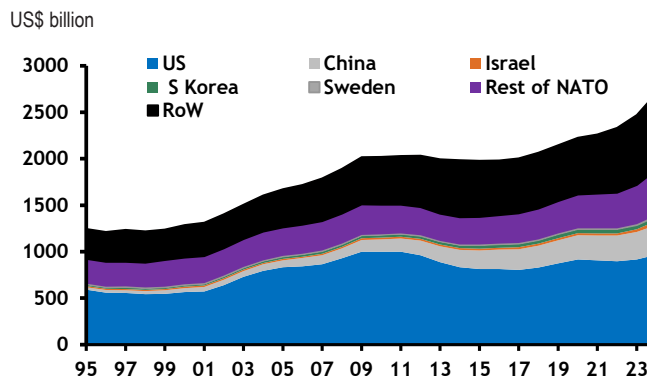
In just the last 20 years, the number of state-based conflicts has increased by 52%, rising from 44 in 2003 at the start of the U.S.–Iraq war to 67 in 2024 (Figure 1). Defense budgets have followed the same trajectory. Global military spending hit a record \$2.7 trillion in 2024, up 34% since 2010 and 16% since 2022, after Russia launched its full-scale invasion of Ukraine. The U.S. accounts for about 36% of global defense spending, while NATO and China make up roughly 18% and 12%, respectively (Figure 2).

Figure 1: State-based conflicts by region



Source: UCDP

Figure 2: Global defense spending



Source: SIPRI

The backbone of U.S. military power lies with its defense industrial base (DIB). It is central to national security, alliance leverage, and military innovation, having evolved through geopolitical shocks, war-driven mobilization, economic restructuring, technological revolutions, and policy choices; in today's era of intensified competition, rapid technology churn, and supply-chain risk, this history explains both current vulnerabilities and the urgency of reform.

From mobilization to consolidation

World War II catalyzed the most ambitious industrial mobilization in American history. With existential threats at the forefront, the U.S. government activated a vast swath of

commercial firms, spanning automotive, electronics, chemical, steel, textiles, and heavy manufacturing, for defense production at unparalleled speed. Engaging more than 800,000 firms (most originally civilian) and creating a highly diversified supplier base with strong resilience and substitutable sourcing, it demonstrated extraordinary surge capacity, exemplified by aircraft output rising from about 6,000 in 1939 to roughly 237,000 in 1944.

During the Cold War, with the Soviet Union as the key military and ideological opponent to the U.S., strategic demands drove greater technological specialization and vertical integration, shifting contracting away from broad commercial manufacturers toward specialized prime contractors; while this deepened expertise and advanced capabilities, it narrowed the supplier base and weakened surge capacity. In his 1961 farewell speech, President Eisenhower famously warned that the rise of the “military-industrial complex” would distort policy, entrench special interests, and erode system flexibility.

The end of the Cold War brought fiscal constraints and strategic uncertainty, leading to calls for efficiency and elimination of redundant capacity. In 1993, then-Deputy Secretary of Defense William Perry hosted an infamous gathering known as the “Last Supper,” urging major industry players to pursue mergers and contractions. Over the next decade, the U.S. defense sector experienced massive consolidation: the number of prime contractors fell from over fifty in 1991 to five by 2000 (Lockheed Martin, Boeing, Northrop Grumman, Raytheon, General Dynamics), with these primes controlling more than 60% of contract value and substantial attrition among Tier 2 and Tier 3 suppliers.

As diversity declined and concentration increased, procurement shifted toward a “flagship-based” doctrine to project overwhelming strength centered on large, technologically integrated platforms (e.g., advanced fighters, carriers, missile systems) that anchor combat capability and alliance interoperability. This approach concentrated resources on marquee programs (such as the F-35 and Ford-class carriers) and deepened integration across weapons, sensors, software, and networks. That architecture produced extraordinary performance in an era when combat advantage could be secured primarily through superior platforms, massed logistics, and uncontested command-and-control.

New threats; new vulnerabilities

But what worked before may not work now. Today’s DIB structure exposes four compounding vulnerabilities:

- **Shallow production depth:** Thinned ranks of tier-2 and tier-3 suppliers create bottlenecks that can stall or cripple programs.
- **Slow acquisition:** Major defense programs drag on for eight years or more, lagging behind rapid technological shifts.
- **Industry concentration:** A handful of dominant primes stifle competition, slow alternative production lines, and weaken system resilience.
- **Supply-chain fragility:** Global, multi-tiered supply chains are efficient but hard to de-risk—especially with dependencies on China for critical materials and components.

These vulnerabilities are strategically acute because modern warfare magnifies their impact. The U.S. now faces a diversified threat landscape: near-peer rivals like China and Russia, mid-tier states wielding cheap precision weapons, and non-state actors exploiting commercial tech. China's Anti-Access/Area Denial (A2AD) and Military-Civil Fusion (MCF) strategies have fused commercial and defense sectors, scaled manufacturing, and overcome chokepoints. China's shipbuilding and missile output dwarfs the U.S., and its supply chain reach—including semiconductors and rare earths—extends globally, even into U.S. defense programs. Other nations (Korea, Sweden, Israel) show the value of tight government-industry integration, supply chain resilience, and rapid adaptation.

Allure and challenge of “assymetric warfare”

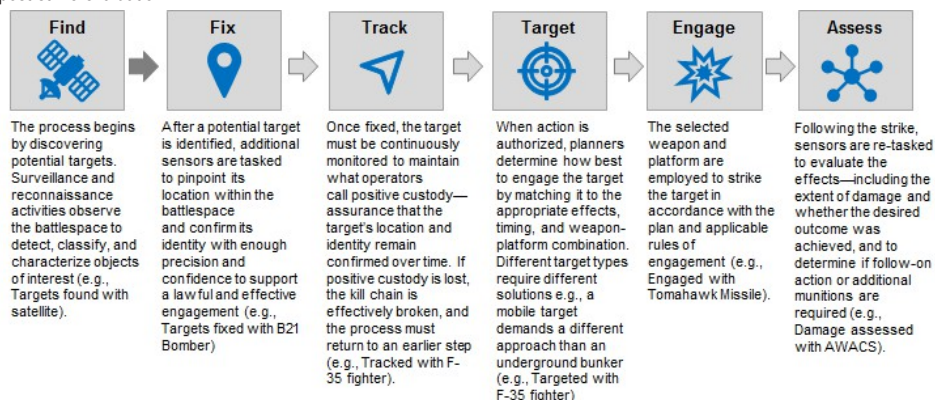
Recent conflicts in Ukraine and Iran have highlighted the need for mass-produced, attritable systems like drones, and the game-changing impact of real-time software and data integration. These developments compress the “kill chain,” enabling faster, more distributed military action and raising tough questions about whether the U.S.—despite its unmatched military power—can both deter and defeat adversaries in a new era of warfare.

Admittedly, these recent wars have increased the allure of asymmetric warfighting, and hence calls for an increased role of large scale, attritable military hardware manufacturing—which the U.S. currently lacks. But a longer term—and just as critical, if not more—lesson is that it has mostly exposed the way in which the digital age has fully permeated into the military realm and the increased role of data technology and software capabilities in shaping the outcomes of future conflicts, or the possibility of avoiding them.

This should not be mistaken for a view that calls for software to replace hardware. It is impossible. Hardware is ultimately always necessary to close what the military refers to as the “kill chain” (Figure 3). Moreover, this should not even be taken as suggesting that attritable hardware should replace the exquisite, powerful, expensive “programs” on which the U.S. has hitherto (and should continue to, in our view) built a good part of its might. The proper way to frame this is that these different layers should complement, enable, and multiply each other's force capabilities rather than substitute each other.

Figure 3: What is a kill chain?

A kill chain is a structured process to detect, identify, and engage a target, then evaluate the results. One commonly used framework breaks this process into six steps: Find, Fix, Track, Target, Engage, and Assess (F2T2EA). The model describes how sensors, platforms, and weapons work together to move from initial detection to a completed strike and post-strike evaluation.



Source: J.P. Morgan, Ultra I&C .

That said, for these layers to act as complements to each other it is necessary that the U.S. defense industrial base undergoes significant changes that span the institutional, the operational and productive/technological—and success is far from granted. The policy impetus has already been set in motion in past years, and builds on the rapid rise of new defense tech companies, with some of them even developing manufacturing capacity at some scale.

However, the policy push still faces several implementation hurdles, which we address in detail below, and institutional roadblocks remain in place. At the same time, there are doubts regarding the capacity of new defense tech companies to move from innovative solutions to production at scale. Last but not least, there are also questions about how large, incumbent defense companies will adjust to these potential changes, whether resisting them, embracing them and adapting, or slowly fading. So far the results suggest that change is taking place but is being slower than some expected, both in terms of the capacity capabilities of new entrants and signs of change in incumbents.

Securing the future

Our broader take, however, is that the direction of travel is likely to be shaped by the following factors. First, as mentioned, the nature of war has changed and communication and control capabilities will be fundamental in the digital age. Second, that successful defense industrial bases elsewhere, each in their own way, have largely based their success on the integration of their defense and commercial industrial bases. Third, and related to the two above, that the U.S. possesses a technological edge in the digital space that sits largely in the commercial sector, and that integrating it to the defense ecosystem is crucial for the U.S. to keep its military dominance.

To maintain military dominance, the U.S. DIB must secure enduring advantages by:

- **Integration:** Forge deeper links between defense and commercial sectors, leveraging shared platforms, open standards, and continuous feedback.
- **Scale:** Build modular, manufacturable systems for rapid mass production, lower barriers for new entrants, and foster a networked industrial base.

- **Innovation:** Drive software-led transformation, iterative hardware development, and commercial digital integration. Emerging tech leaders are reshaping autonomy, hypersonics, advanced manufacturing, satellite communications, and mission resilience.
- **Policy reforms** like NDAA and Modular Open Systems Architecture (MOSA) are promising, but must overcome entrenched acquisition processes, PPBE constraints, and persistent friction over intellectual property and data rights—barriers that can either enable competition and modularity or reinforce vendor lock-in and exclude new entrants.

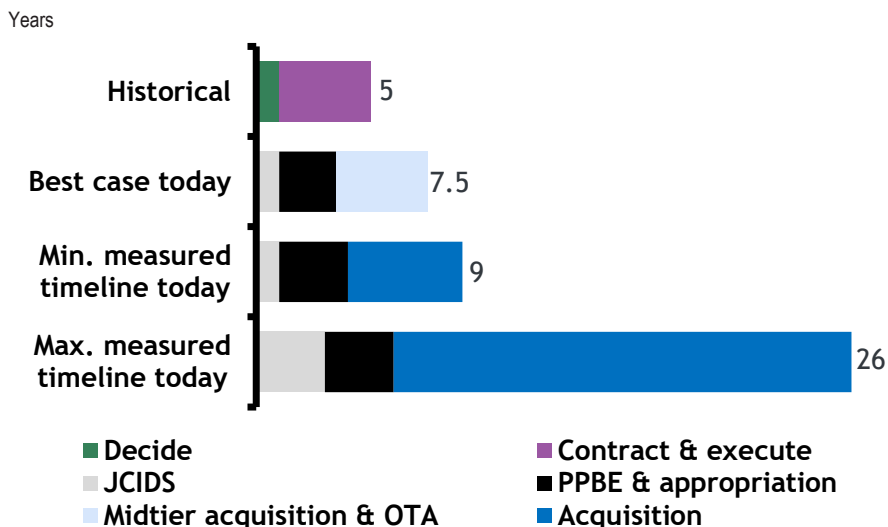
America's defense industrial base stands at a crossroads. Adapting to persistent competition, rapid technological change, and mass-scale attrition warfare is not optional; it's essential. Enduring advantage will depend on integrating commercial and defense sectors, scaling production, and accelerating innovation, while dismantling deep-seated institutional and structural barriers. Failure to act risks eroding deterrence and long-term military effectiveness, with consequences for national security and global leadership.

The U.S. defense industrial base

Since the end of the Cold War, the U.S. defense industrial base (DIB) has been acclaimed for its ability to deliver world-class, high-end platforms such as aircraft carriers, stealth aircraft, and advanced missile systems, which underpin its global military dominance (see Appendix: I. U.S. and Chinese Military Capabilities). These achievements, however, rest atop a deeply specialized and concentrated network of contractors dominated by a handful of incumbent "primes." This underlying architecture has become brittle, exposing the system to several vulnerabilities.

One telling metric that helps sum up many of the issues is the lengthy timelines for developing major defense acquisition programs, averaging eight years, which means that 70% of the electronics procured by the defense forces are obsolete before they even reach the field (Figure 4).

Figure 4: DoD's procurement process timeline



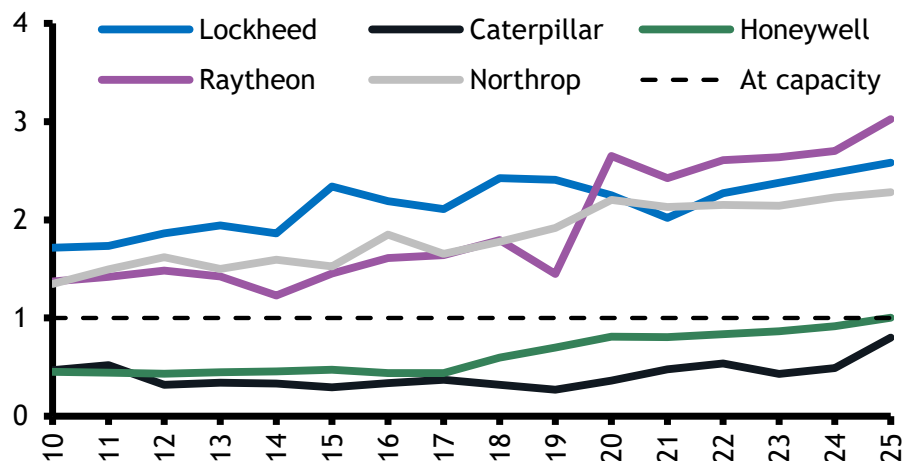
Source: Hudson Institute and J.P. Morgan

More broadly, there are a handful of overarching issues with the current DIB that have been widely noted.

- **First, the production base**—especially among tier-2 and tier-3 suppliers—has thinned dramatically. The specialized segments that produce castings, energy materials, advanced electronics, propulsion components, and critical tooling are narrow, often geographically concentrated, and capacity-constrained. This creates hidden chokepoints that can halt or slow even fully funded programs, as a shortage in a single critical component or manufacturing line cascades through the entire system. Such bottlenecks reflect not just the prime's ability to assemble, but an upstream fragility in the broader industrial ecosystem.
- **Second, foreign dependency.** Even where final assembly occurs domestically, the U.S. defense supply chain is tightly interwoven with global industrial networks, and in many instances is directly exposed to dependencies linked to China. Raw materials, processed inputs, and specialized manufacturing steps are frequently sourced abroad. These dependencies create acute risks: strategic disruption becomes plausible, either through geopolitical coercion, export controls, market manipulation, or simple supply chain shocks. Substituting these foreign supply sources is not a simple fix; it requires recertification, qualification, and navigating the same slow-moving acquisition logic that created the vulnerabilities to begin with.
- **Third, the production throughput among the primes.** Strained book-to-bill ratios signal that the existing defense industrial base is consistently operating under strain, which is not the case across the broader commercial industrial base (Figure 5). Limited labor pools, single-point suppliers, and the need for long lead items translate into rising backlogs that indicate demand is outpacing available productive capacity. In a system optimized for reliability and predictability rather than volatility and surge, this is a warning sign: “surge” is not a switch to be flipped, but a capability that must be cultivated and maintained.

Figure 5: Capacity constraints notable among the primes

Book-to-bill ratio



Source: Bloomberg Finance L.P. and J.P. Morgan

Compounding these structural frailties is industry concentration. Decades of consolidation among primes and key suppliers has eroded competitiveness. Barriers to entry remain steep due to capital requirements, compliance burdens, and incumbency advantages in navigating complex military requirements and sustainment. As a result, industry incentives often favor protecting existing program positions and margin structures, rather than cost-driven innovation or manufacturing efficiency. Limited competition also slows the emergence of alternative production lines, depriving the system of vital resilience; a supplier shock rapidly becomes systemic.

Box 1: A Schematic of the U.S. DIB's Supply Chain

The DIB's supply chain is a multi-tiered, international network comprising a diverse array of contractors, all dedicated to equipping the military with the advanced systems necessary to achieve its strategic objectives.

This network encompasses thousands of companies, ranging from major "prime" (Tier 1) contractors—such as Lockheed Martin, Raytheon, Boeing, and Northrop Grumman—to small and medium-sized enterprises that provide highly specialized components (Figure 6). Importantly, the DIB's scope extends well beyond hardware, incorporating critical domains such as software development, cybersecurity, and logistics services.

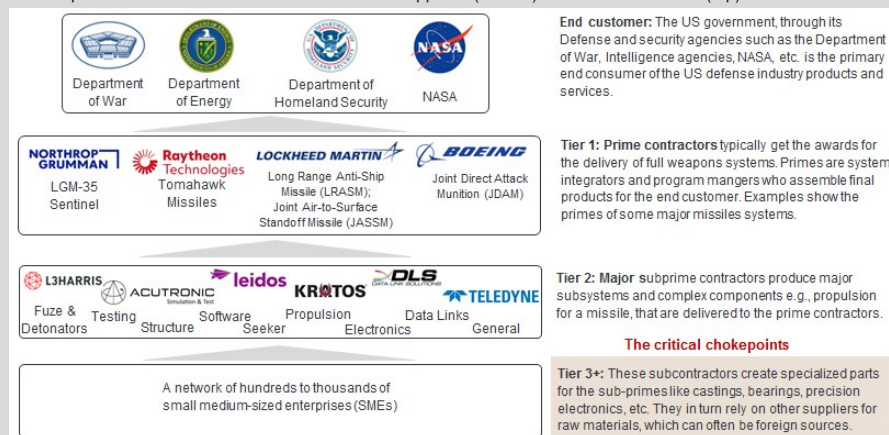
A defining characteristic of the DIB is its monopsonistic structure: while the sector is largely driven by private enterprise, the U.S. government—primarily the Department of War—serves as the principal buyer, shaping market demand through the National Defense Strategy (NDS). The Department of War typically awards contracts for complete weapon systems to prime (Tier 1) contractors. For instance, in 2025, Lockheed Martin secured a \$12.5 billion contract to deliver 300 F-35 fighter jets to the U.S. Navy.

Prime contractors, in turn, depend on an extensive network of subcontractors (Tier 2 and beyond) to supply essential subsystems—such as propulsion, weapons, and

navigation systems—that are integrated into the final product. In many cases, Tier 2 suppliers, such as those providing weapons systems, further rely on Tier 3 and lower-tier contractors for critical components like fire control systems and warheads. This intricate, layered structure of subcontracting is fundamental to the delivery of defense products, yet it has also posed persistent challenges to the DIB for decades.

Figure 6: U.S. Defense Industrial Base Supply Chain

Flow of products and services from raw materials suppliers (bottom) to the end-customer (top)



Suppliers shown are not exhaustive and are for illustrative purposes only
Source: War on the Rocks, Govini, JP Morgan

Warfare is outpacing industry

The vulnerabilities afflicting the traditional defense industrial base are not static; they are being compounded by the accelerating evolution of warfare itself. Since the early 2000s, combat has grown more transparent at the tactical level, fueled by advances in sensors, drones, and commercial satellite imagery, as well as distributed reconnaissance and targeting. The Ukraine conflict has massively accelerated these trends, making clear that operational advantage now flows from the ability to connect sensors and shooters, maintain decision superiority under attack, and rapidly regenerate combat power despite attrition.

Additionally, U.S. forces face increasing challenges from adversaries pursuing Anti-Access/Area-Denial (A2/AD) strategies, particularly exemplified by China. These adversaries contest forward bases, threaten concentrated formations, and deny access to critical maritime and air corridors, pushing U.S. doctrine toward distributed operations, longer-range effects, and dynamic force employment. In other words, this is a strategy designed to make it hard for an opponent to get into a region (“anti-access”) and hard for them to operate effectively once they’re there (“area denial”). It’s a playbook for turning a contested zone into a place where enemy forces can’t safely arrive, can’t communicate well, can’t see clearly, and can’t move or fight freely.

It typically combines tools like electronic warfare (jamming), cyber attacks, long-range missiles, and advanced air defenses to disrupt communications and sensing, threaten ships and aircraft from far away, and raise the cost and risk of operating near the contested area. This evolving environment punishes rigid architectures and demands that systems be plug-and-play, reconfigurable, and able to operate under degraded

connectivity.

At the center of the response to this paradigm shift is the U.S. pursuit of Joint All-Domain Command and Control (JADC2), which seeks to connect sensors, decision-makers, and weapons across land, sea, air, space, and cyber domains, with systems such as Palantir's Maven (MSS) and Anduril's Lattice playing a central role^{1,2}. The DoD uses Uber as an analogy for JADC2, suggesting that instead of a rider being matched to the best nearby driver, the military aims to connect sensors from every service (Army, Navy, Air Force, etc.) into a single network and use software (including AI) to rapidly match an emerging target or threat to the best available shooter or effect—whether a missile, aircraft, cyber tool, or electronic warfare option—based on factors like proximity, timing, and suitability, enabling much faster decisions than today's slower command cycles.

JADC2 is not merely a technical upgrade; it represents a fundamental transformation—enabling a common operational picture that is actionable, resilient, and able to route targeting and tasking decisions through robust pathways even when under attack. Victory in modern contested environments depends on keeping the sense-decide-act loop functioning faster and more reliably than the adversary.

The existing industrial base, optimized for long development cycles and stable sustainment, struggles to meet these requirements. The decisive variables are now the cadence of software updates, integration of new sensors, rapid fielding of mission applications, and replenishment of large numbers of attritable assets, not merely the delivery and maintenance of exquisite platforms. Requirements have shifted from “build once, maintain forever” to “build, iterate, and continuously integrate.”

Taken together, the traditional U.S. defense industrial base is capable of producing excellence, but struggles to generate at scale, rapidly iterate at operational speed, or robustly de-risk its supply chain amid geopolitical disruption. These weaknesses predate current shifts in warfare but are becoming exponentially more dangerous as the operational environment demands speed, dispersion, and constant adaptation.

This new reality demands not just an upgrade to the traditional defense industrial base, but a fundamentally different model: one that mirrors the commercial technology ecosystem in its integration patterns, iteration speed, and ability to scale production rapidly. Integration between the defense and commercial industrial bases is now a precondition for the adaptability and resilience required by modern warfare. As we analyze below, this is a common feature across different successful DIBs elsewhere, regardless of their disparate structures and objectives.

Lessons from Other DIBs

China's Military-Civil Fusion

The U.S. Department of War (DoW) identifies China as posing the biggest competition and threat to the security of the U.S. DIB. China's defense industrial base has grown as

1. Hoehn, John R. Joint All-Domain Command and Control: Background and Issues for Congress. Congressional Research Service, 21 Jan. 2022

2. <https://investors.palantir.com/news-details/2024/Palantir-Expands-Maven-Smart-System-AIML-Capabilities-to-Military-Services/>

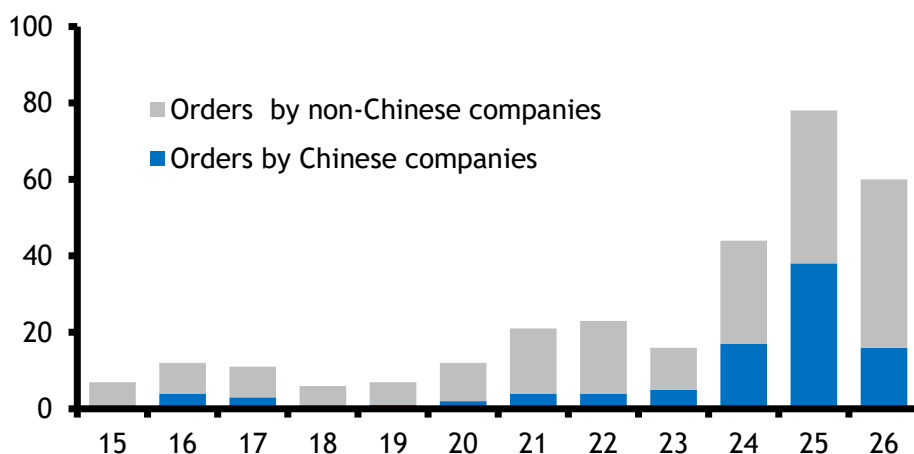
it has become a global economic force over the last couple of decades. During this time, China has pursued policies that have allowed it to overcome many of the kinds of challenges and chokepoints that the U.S. defense industrial base currently faces. China's primary advantage lies in its Military-Civil Fusion (MCF) strategy. Unlike the U.S., where the commercial and defense sectors are largely distinct with a buyer-vendor relationship, China's DIB is physically and operationally entwined with its global manufacturing base.

China's MCF promotes deep integration between its defense and civilian industries. The MCF ensures that new innovations simultaneously advance economic and military development by removing barriers between China's civilian and commercial sectors and its military and defense industrial sectors. For example, commercially available Chinese DJI drones have been deployed in the Ukraine-Russia war with great effectiveness. The US DoW maintains a list of Chinese companies that contribute to the MCF or may have other connections to the Chinese military; as of 2026, the US DoW has added 13 Chinese drone companies, including DJI. Designation prevents companies from participating in DOD procurement³.

Another notable example that illustrates China's use of MCF is the Chinese People's Liberation Army (PLA) use of commercial cargo vessels to conduct amphibious training drills around the Taiwan Strait in August 2022, with modified civilian Roll-on/Roll-off (RoRo) cargo vessels from companies such as China's COSCO Shipping Lines. Reports indicate that China has accelerated construction of these RoRo vessels because of their dual military-civilian capabilities (Figure 7). Commercially, Chinese automakers are increasingly in need of RoRo vessels to transport EVs overseas, as Chinese Battery EV sales grew 1.2x in 2021-2023⁴.

Figure 7: RoRo shipbuilding orders at Chinese shipyards

of orders based on delivery year



Source: CSIS China Power Project, New Ship Orderbook, China Shipbuilding, J.P. Morgan.

3. <https://www.dji.com/media-center/announcements/dji-statement-us-court-ruling-regarding-department-of-defense>

4. Matthew Funaiole, Brian Hart, Jaehyun Han, Jennifer Jun (2023), "China accelerates construction of 'Ro-Ro' vessels, with potential military implications" CSIS

While concerns about MCF have only recently gained attention in the U.S., China has been pursuing MCF since the early 1980s. This has not only been a consistent objective of Chinese leaders, but also illustrates that the structural changes required to realize this policy took a long time to occur even for a strong centrally controlled economy. Dedicated funding programs for the MCF enabled the cross-pollination of technology. For example, from 2015-2021, China established over 35 funds and dedicated about \$70B to investing in firms related to the MCF according to estimates ⁵. Additionally, according to the DoW, China has taken unprecedented steps to facilitate the entry of private firms into its defense industrial base. An assessment identified over 180 private Chinese companies that have contributed goods & services to China's defense aviation and aerospace⁶.

Furthermore, China's research, development, and acquisition (RDA) process has improved over the last decade. Similar to the U.S., the process generally takes 10-15 years for major platforms but is highly iterative. Moreover, other analysis note that priority projects such as hypersonic missiles can be accelerated and it takes China less than seven years to deliver an operational capability from an idea, compared to an average of 16 years for the U.S. ⁷. A major factor that gives the Chinese RDA process an efficiency advantage over that of the U.S. is that the major Chinese defense firms are State Owned Enterprises (SOEs) which are closely controlled by the CCP ⁸. For example, five of the top 25 defense companies globally by arms revenue are state-owned by the CCP (See Appendix: V. Main global public defense companies* & VI. Main U.S. private defense companies).

How China leverages commercial manufacturing

China's expansive and technologically advanced manufacturing sector sits as the bedrock of its DIB, facilitating the rapid development, prototyping, production, and deployment of military systems. Today, China commands approximately 30% of global manufacturing output—nearly double that of the United States. This dominance has been achieved through a suite of policy instruments, including low-interest loans, subsidized tax rates, relaxed environmental regulations, and strategic dumping practices, all designed to bolster industrial growth ([Securing rare earth supply: Key is technology](#)).

The sheer scale of China's manufacturing capacity has enabled the swift expansion of its military assets. Notably, China has emerged as the world's largest shipbuilder, with a shipbuilding capacity more than 230 times greater than that of the U.S. ([Revitalizing U.S. Shipbuilding: Automation, autonomous, partnerships, and shipping](#)). Individual shipyards, such as Jiangnan, possess more capacity than all U.S. shipyards combined ⁹. The Jiangsu shipyard, for example, increased its production of RoRo vessels eightfold between 2015 and 2026 (Figure 8).

5. Elsa Kania and Lorand Laskai (2021), "Myths and Realities of China's Military-Civil Fusion Strategy", Center for a New American Security

6. Seth Jones, Alexander Palmer, (2024), "Rebuilding the Arsenal of Democracy – The US and Chinese Defense Industrial Bases in an Era of Great Power Competition", CSIS

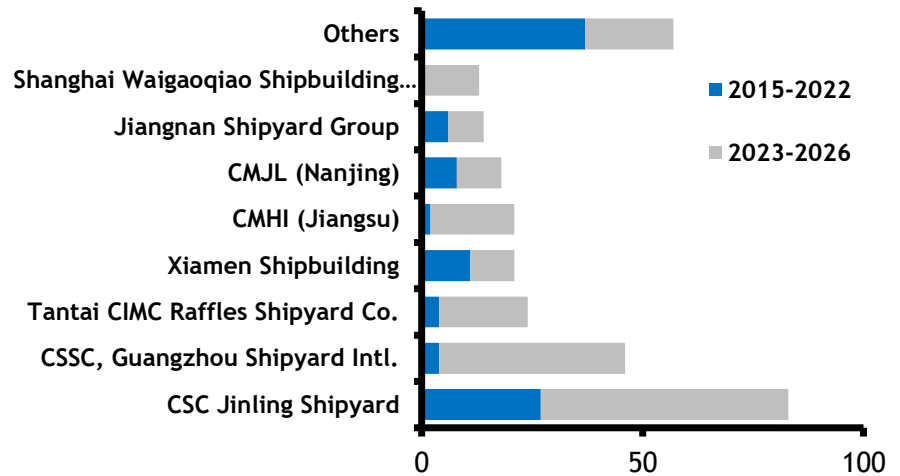
7. Dan Patt and William Greenwalt (2021), "Competing in Time: Ensuring Capability Advantage and Mission Success through Adaptive Resource Allocation", Hudson Institute

8. Christian Curriden (2023), "The Chinese Acquisition Process", RAND Corporation

9. Seth Jones, Alexander Palmer, (2024), "Rebuilding the Arsenal of Democracy – The US and Chinese Defense Industrial Bases in an Era of Great Power Competition", CSI

Figure 8: Chinese shipyards capacity

Number of ships

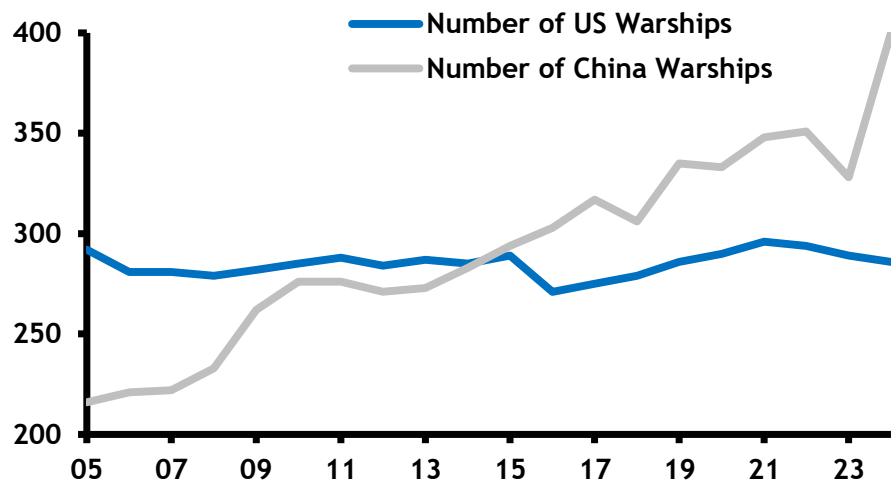


Source: CSIS China Power Project, New Ship Orderbook, China Shipbuilding, J.P. Morgan.

This industrial prowess confers a significant strategic advantage in the event of a protracted maritime conflict with a near-peer adversary such as the U.S. As a result, the People's Liberation Army Navy (PLAN) has ascended to become the largest navy in the world by hull count, with an estimated 400 warships in 2025, approximately 40% more than the U.S. Navy's fleet (Figure 9). China's manufacturing scale and agility not only underpins its military industry's strength, but also positions it favorably for sustained operations across multiple domains.

Figure 9: U.S. and Chinese naval warships

Units

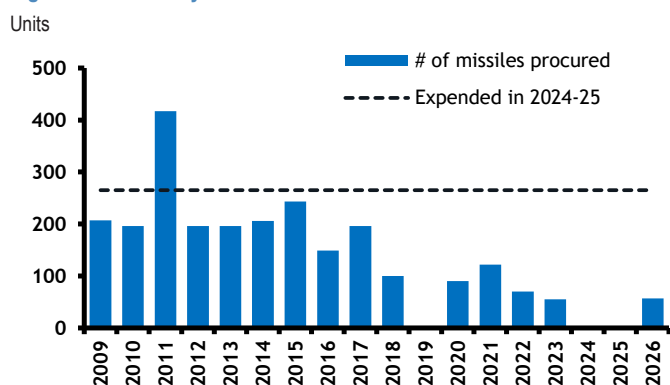


Source: Congressional Research Service

China's manufacturing scale has also conferred advantages in the expansion of its missile programs. According to a Department of War report, China now possesses the most active and diverse ballistic missile program in the world, with a rapidly growing

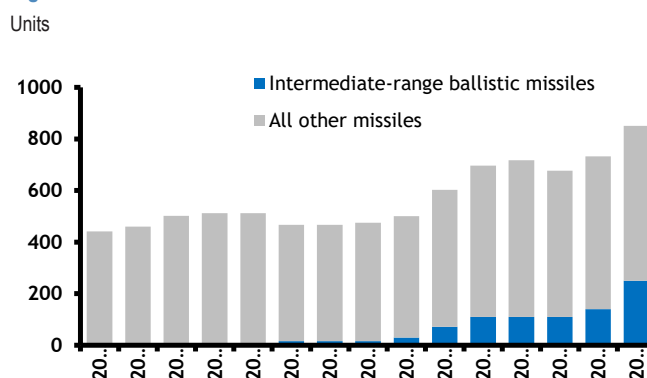
stockpile and an exceptional rate of production, including hypersonic glide vehicles ¹⁰. China conducts more ballistic missile tests than the rest of the world combined, underscoring its commitment to technological advancement and operational readiness. The DoW estimates that the number of active intercontinental ballistic missiles (ICBMs) in China has increased by approximately 200% between 2016 and 2022 - illustrated by the number of missile launchers. To illustrate the scale of the increase, U.S. Navy procurement of Tomahawk missiles rose by 44% over the same period, assuming no missile expenditures (Figure 10 & Figure 11). More importantly, the current conflict between the U.S. and Israel versus Iran has highlighted the challenges faced by the U.S. DIB in manufacturing and the impact on munitions supply.

Figure 10: U.S. Navy Tomahawk missiles



Source: Bloomberg Finance L.P., J.P. Morgan.

Figure 11: Chinese missile launchers



Source: IISS, J.P. Morgan.

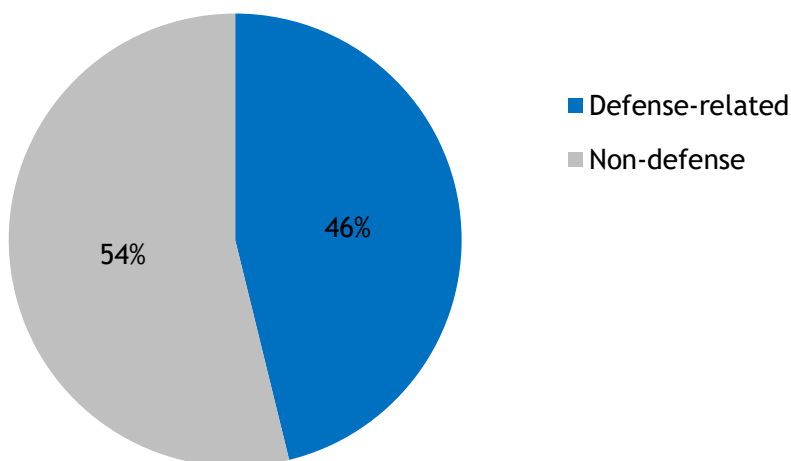
Recent media reports, supported by satellite imagery, reveal that 65 out of 99 sites associated with missile manufacturing have expanded, enabling China to further increase production ¹¹. This growth has been facilitated by the consolidation of major state-owned missile producers, such as CASC, which has acquired smaller companies and research institutes, centralizing the development and manufacturing of missiles, engines, and launch systems. In contrast to the U.S. defense industrial base, where critical suppliers have shuttered operations due to international competition and inconsistent demand from the DoD, Chinese suppliers have successfully tapped into global markets. Over the past decade, China's defense exports have surged by 53%, with integrated circuits alongside other components that are part of the defense supply chain accounting for up to 46% of exports in 2024 (Figure 12). This robust export activity has sustained China's manufacturing capacity well beyond domestic demand.

10.Office of the Secretary of Defense, Military and Security Developments Involving the People's Republic of China, 2022, 149

11.Tamara Qlblawl, Glanluca Mezzofiore, Thomas Bordeaux, Yong Xlong, Lou Robinson (2025), "A new arms race: Satellite images, maps and records reveal huge surge in China's missile production sites", CNN

Figure 12: China's exports have a large defense-related component

% of major goods exports



Source: NBSC, J.P. Morgan

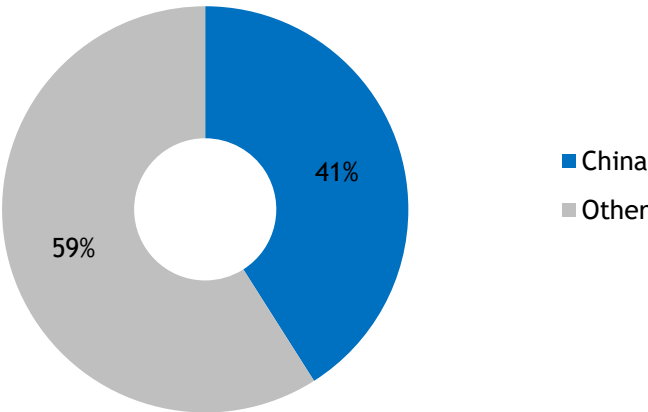
China's supply chain network

China's defense industry is strengthened by its global dominance in the supply chain for strategic materials and components. As the world's leading producer of rare earth elements, essential for advanced weapons systems, China supplies up to 90% of global rare earths and has leveraged this market position to influence both supply and pricing ([Securing rare earth supply: Key is technology](#)). This advantage not only supports China's own defense industrial base, but also extends its reach into the supply chains of other nations, including the U.S. A 2023 analysis by Govini revealed that between 2005 and 2023, the number of Chinese firms embedded in U.S. supply chains nearly tripled to about 43,000 ¹². Several factors have contributed to this surge: the proliferation of consumer electronics such as the iPhone and e-commerce, the full integration of Chinese firms following China's accession to the WTO in 2001, and most importantly the widespread offshoring of production to China by Western companies seeking cost efficiencies ([FDI: Shifting from outsourcing to securing supply chains](#)).

The implications for U.S. defense are significant. Some estimates suggest that as much as 41% of the Department of War's weapons systems rely, at least in part, on Chinese semiconductors (Figure 13). Prime contractors for key missile systems—including Tomahawk, LRASM, JDAM, and JASSM, such as General Dynamics, Raytheon, Boeing, Northrop Grumman, Lockheed Martin, depend on hundreds of subcontractors for critical components like propulsion systems, electronics, test equipment, detonators, and data links, who in turn depend on hundreds of suppliers. Govini's analysis identified as many as 1,587 Chinese firms supplying materials to these subcontractors. This reliance on Chinese suppliers poses a substantial risk to the resilience of the U.S. defense industrial base, particularly in the context of a protracted conflict.

12.Jeb Nadaner, "Numbers Matter: Defense Acquisition, U.S. Production Capacity, and Deterring China", Govini

Figure 13: DoD weapons and infrastructure supply chain that relies on Chinese semiconductors
%

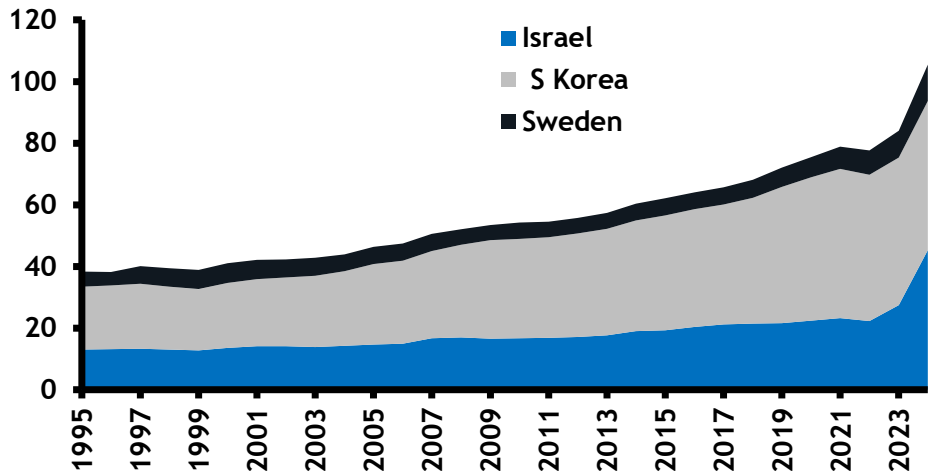


Source: Govini, 2023

Other DIBs: Korea, Sweden, Israel

While the U.S. remains the dominant global defense player, particularly in innovation, and China’s production scale remains unmatched, a significant shift in cost competitiveness and innovation speed is occurring in the defense industrial bases of other countries, illustrated by Israel, Sweden and Korea. Their defense industrial bases are shaped by their unique geopolitical environments, technological advancements, industrial production, and models of government-industry collaboration. Each of these allied nations has forged a globally competitive defense sector, but their operating models, leading companies, and strategic strengths and weaknesses differ in ways that offer valuable insights for the U.S. DIB.

Figure 14: Defense spending by Israel, South Korea and Sweden
US\$ billion



Source: SIPRI, J.P. Morgan

Korea - The large exporter

South Korea's defense budget has increased an average of 53% from \$32B in 2010 to \$49B in 2024, shown in Figure 14¹³. At the same time, arms sales have been increasing over the last decade. Total arms and ammunition exports grew an average of 10x from \$245M in 2010 to \$2.7B in 2024¹⁴. South Korea's rapid innovation cycles and advanced manufacturing capabilities have enabled it to become a major supplier of land systems, naval assets, and missile technology. The industry produces a wide range of weapons, covering conventional weapons and advanced technology-oriented weapon systems across the ground, naval, aerospace, and electronic equipment fields¹⁵. The defense industrial base is based on a joint-venture model between government agencies and private companies.

The Ministry of National Defense and the Defense Acquisition Program Administration (DAPA) set strategic priorities and fund research and development, while weapons are produced by 6 main companies: Hanhwa Aerospace, Korean Aerospace Industries, Hyundai Rotem, LIG Nex1, Hyundai Heavy Industry, and Poongsan. In this Public-Private Partnership, the government undertakes the conceptualization, design, experimentation, and testing of a weapon system, while private defense companies undertake mass production to provide complete products for the armed forces and support aftersales maintenance for decades.

New weapons development begins with a government-commissioned R&D project, which is carried out by a government research institute, the Agency for Defense Development (ADD), rather than private companies – which contrasts with the U.S. process in which industry carries out R&D. Korea's ecosystem, however, has the advantage that it spares enormous R&D expenses and has more financial flexibility to compete in and out of the Korean market¹⁶.

The DAPA oversees weapons procurement and is the single point of contact for weapons trade. Analysts have identified several factors that drove the fast growth in the Korean DIB, including (1) its continuous military tension with North Korea, (2) its international partnership with the U.S. and other major defense powers that provided S. Korea with weapons and technology transfers that eventually became the foundations of Korea's own products, and (3) Korea's bipartisan political support for the defense industry. Finally, the Russia-Ukraine war boosted weapons exports to Europe (Figure 15), e.g., the 2022 sale of K2 Black Panther tanks to Poland, one of the largest defense deals in South Korean history, illustrates both the country's export success and its dependence on foreign markets¹⁷.

13 SIPRI

14 Korea Customs Service

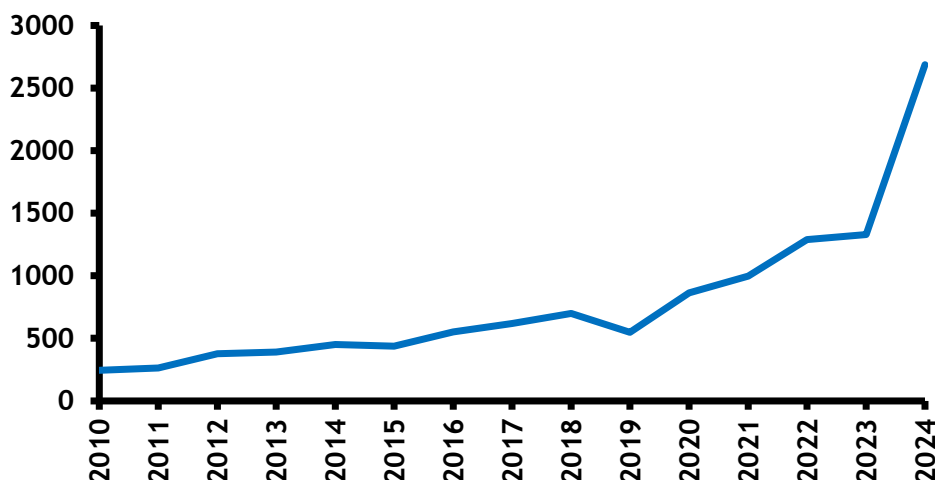
15 Wooyeal PAIK (2024), "South Korea's emergence as a defense industrial powerhouse", Center for Asian Studies and Securities Studies Center

16 Wooyeal PAIK (2024), "South Korea's emergence as a defense industrial powerhouse", Center for Asian Studies and Securities Studies Center

17 Daniel Darling (2022), "Poland agrees to massive arms package with South Korea", Defense and Security Monitor

Figure 15: S. Korea arms and ammunition exports

US\$ million



Source: South Korea Customs Service, J.P. Morgan

Sweden - Reliable supply chains

The Swedish defense industry has experienced remarkable growth and transformation, particularly following Sweden's accession to NATO in March 2024. The nation's defense budget has more than doubled over the past decade, rising from \$5.3 billion in 2010 to \$11.7 billion in 2024, with a striking 34% increase between 2023 and 2024 alone (Figure 14). Despite its modest size, Sweden's defense sector is highly specialized and globally competitive, anchored by prominent firms such as Saab AB and BAE Systems Hägglunds. Sweden has emerged as a significant arms exporter, especially within Europe. Notably, 60-70% of Saab's profits are derived from exports, as highlighted by CEO Micael Johansson ¹⁸. The industry's focus on niche technologies—including electronic warfare, stealth, and advanced sensors—enables Sweden to excel on the international stage. Saab's development of the A26 submarine, the world's first 5th generation submarine per Saab, exemplifies the country's reputation for innovation ¹⁹.

While Saab and BAE Systems Hägglunds are the industry's major players, Sweden's defense ecosystem is enriched by a multitude of technology companies producing dual-use products for both civilian and military applications. Approximately 400 companies supply the Swedish Defense Materiel Administration (FMV), the government agency responsible for procuring, developing, and supplying equipment and services to the armed forces. The Swedish Security and Defense Industry Association (SOFF) represents industry partners and collaborates closely with FMV to support national defense objectives ²⁰. FMV's partnership with industry ensures that procurement and research and development are closely aligned with Sweden's defense needs, similar to the approach seen in the U.S. Department of Defense.

¹⁸Stew Magnuson (2024), "Sweden Brings Defense Industrial Might to NATO", National Defense

¹⁹<https://www.saab.com/products/5th-generation-submarine>

²⁰Defense industry strategy for a stronger Sweden (June 2025)

A defining characteristic of the Swedish defense industry, rooted in Cold War ambitions, was the pursuit of independence from foreign suppliers. However, the contemporary landscape reflects a growing need for material cooperation with allies, and Sweden now relies on deliveries of components and systems from abroad ²¹. For example, Bloomberg supply chain data indicates that 70% of Saab's direct suppliers are domiciled in the following countries: U.S. (21%), Sweden (16%), United Kingdom (13%), France (11%), and Germany (9%). Notably, none of the direct suppliers is domiciled in China even though about 4% of all direct supplier facilities are located in China.

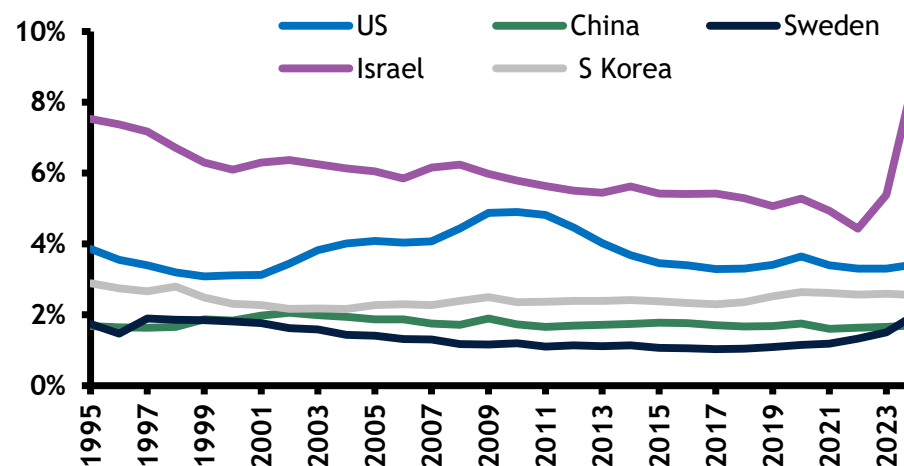
Ultimately, Sweden's small domestic market and limited production capacity necessitate a strong reliance on exports and international partnerships. The nation's ability to innovate and specialize has allowed it to carve out a prominent role in the global defense industry and is actively adapting to the changing global conflict environment.

Israel - Tech-driven innovation

Israel's defense industrial base is a model of adaptability, rapid innovation, and export-driven growth, underpinned by strong government-industry collaboration and integration with the tech sector. With one of the highest defense-to-GDP ratios globally, Israel's defense budget has more than doubled over the past decade, rising from \$16.7B in 2010 to \$45.3B in 2024, with a striking 65% increase between 2023 and 2024 alone following the October 7th, 2023, attacks by Hamas (Figure 16) ²².

Figure 16: Defense spending by country

% of GDP



Source: SIPRI, J.P. Morgan

The Israeli DIB has a multi-tiered, structured network similar to the U.S. DIB in that a handful of large companies develop and manufacture major weapons systems and components. These companies include Israel Aerospace Industries (IAI) and Rafael, which are both state-owned, and Elbit Systems, which is privately owned. Elbit is the largest defense company in Israel after acquiring several private companies in the 1990s. Sitting below these firms are several hundred second-tier plus companies that produce

²¹Robert Limmergard and Jonathan Lindqvist (2023), "Swedish Defense-industrial capabilities shine bright", European Security & Defense

²²SIPRI

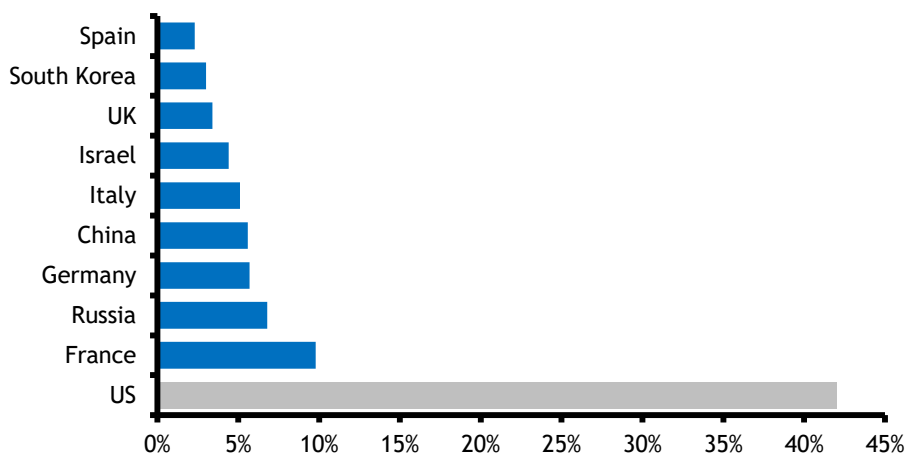
and supply subsystems, components, and services used by the major companies. The government plays a central role in shaping the defense industry, mainly through its civil-military fusion, very similar to the Chinese model. Israel's defense sector is deeply integrated with the civilian high-tech industry, startups, and academia, which fosters dual-use technology development and ensures that new solutions are quickly adapted to evolving threats.

Despite its small domestic market, Israel is one of the largest weapons exporters globally with a 4% global market share (Figure 17). IAI, Rafael, and Elbit account for over 70% of Israel's defense exports²³. For example, Israel's Iron Dome missile defense system, developed by Rafael, has intercepted thousands of rockets and is now being exported to countries including the US and South Korea. The defense industry's focus on specialized high value products such as cyber systems, UAVs, C3, and missiles, combined with its agility to pivot in response to changing market conditions, security needs, and technological trends have accelerated its global expansion.

Moreover, Israeli defense firms have established overseas subsidiaries and partnerships, enabling access to foreign markets and also to integrate their technologies into larger, multinational defense programs. For example, these subsidiaries also enable Israeli companies to supply products to the Israel Defense Forces (IDF) using U.S. Foreign Military Sales (FMS) funding, which often mandates that a portion of military aid be spent on U.S.-based suppliers or through U.S.-based entities.

Figure 17: Market share of leading arms exporters

% of global exports, 2021-25



Source: SIPRI, J.P. Morgan

Defense scholars have put forth several explanations for how Israel's DIB evolved into a world class industry despite challenges such as a small local market, arms embargoes in the 1960s by France – its primary supplier at the time – and limited raw materials. These explanations include Israel's approach to counter adversaries' numerical superiority with innovation, substantial financial support from and technological collaboration with the U.S., access to a highly skilled local workforce and scientific infrastructure, adaptability to changing local and global arms markets, and increasing collaboration in arms

23.Yoram Evron (2024), "Israel's defense industry: adaptation and growth in a changing arms market", Defense Studies, 25(2), 301-32

development with the civilian sector.

The lessons

Across peer and partner defense models, one clear pattern emerges: regardless of political or military differences, defense effectiveness increasingly relies on tight integration between commercial and defense industrial bases. This integration accelerates innovation, production, and operational learning. China's military-civil fusion achieves this at scale, intertwining defense and manufacturing to rapidly convert civilian capacity into military output and move dual-use technologies into operational use.

Supply chain resiliency is another critical lesson. It's not just a supporting factor—it's essential for wartime endurance. Sustained production and replenishment depend on minimizing single points of failure, foreign dependencies, and upstream bottlenecks. China pursues resiliency through vertical integration and industrial sovereignty, controlling key production steps and materials. Sweden, by contrast, builds resiliency through trusted alliances and partner supply networks, reducing vulnerability by relying on reliable partners rather than adversaries.

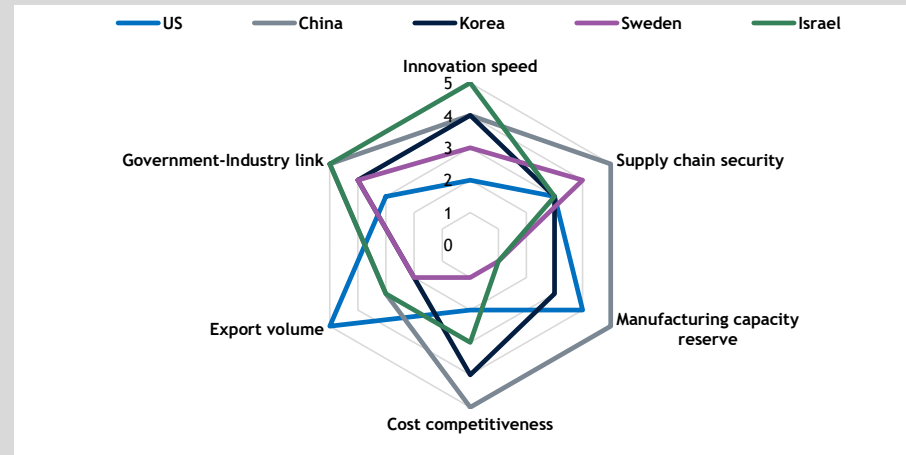
Finally, while technology and innovation are central, their impact depends on the ability to quickly translate operational needs into fielded capabilities. Israel exemplifies this approach, with its defense sector tightly linked to the civilian high-tech ecosystem, startups, and academia, enabling rapid adaptation of dual-use tools and shortening the feedback loop from combat experience to deployment.

Box 2: Comparative Analysis of DIBs

To summarize some of the findings in the prior section, below we present a comparative analysis of the U.S. DIB and those of adversaries and peers. The analysis reveals the relative strengths and weaknesses of each DIB model. Of note, this analysis refers to features that contribute to a strong, resilient and dynamic defense industrial base, not military might or capability. We identify six features that have driven the success of the various DIBs to qualitatively illustrate how/where U.S. defense companies and the DoD could improve the resilience and security of the U.S. DIB. Figure 18 summarizes our analysis, showing how different DIBs rank on each of six features: Innovation Speed, Supply Chain Security, Manufacturing Capacity Reserve, Cost Competitiveness, Export Volume, and Government-Industry Links. A detailed analysis to determine the score rankings is available upon request.

Figure 18: Comparative analysis of U.S. and other DIBs

Score, 1= worse, 5= best



Source: J.P. Morgan

(i) Innovation Speed: This measures how quickly a military need is identified, prototyped, and deployed as a frontline capability—the so-called “combat loop.” Israel leads globally, with the IDF able to field new technologies in months, far outpacing U.S. and European acquisition timelines. Israeli startups can push software patches or hardware updates in days, a feat hampered in the U.S. by procurement bureaucracy²⁴.. Unlike some Western tech firms, Israeli high-tech companies actively embrace defense projects, often enhancing their reputation by doing so ²⁵..

(ii) Supply Chain Security: This refers to the reliability and sovereign control a country’s DIB has over its entire value chain—from raw materials to final assembly and maintenance. China excels here, achieving vertical integration and industrial sovereignty through its Military-Civil Fusion strategy, with the PLA controlling every step via state-owned defense companies. The U.S., by contrast, faces manufacturing chokepoints and relies on China for critical minerals.

(iii) Manufacturing Capacity Reserve: This is the latent ability to mass produce, sustain, and surge military hardware during peace and conflict. China dominates, with ship production capacity 230 times greater than the U.S., and the ability to rapidly convert civilian factories for military use. The U.S. defense industry is often hesitant to invest in capacity expansion due to inconsistent demand signals, but the broader U.S. industrial sector holds significant untapped potential.

(iv) Cost Competitiveness: This measures the ability to produce and sustain military hardware at lower costs without sacrificing performance. China’s low manufacturing and labor costs give it a major advantage, while U.S. systems are typically more expensive due to their sophistication. Cost competitiveness is shifting global defense strategies to prioritize quantity over producing the most exquisite and sophisticated systems. Israel, for example, designs weapons for mass production to keep costs low and scale flexibly.

24.Aosheng Pusztaszeri and Emily Harding (2025), “Technological Evolution on the Battlefield”, CSIS

25.Yoram Evron (2024), “Israel’s defense industry: adaptation and growth in a changing arms market”, Defense Studies, 25(2), 301-32

(v) Export Volume: The U.S. is the world's top arms exporter, holding a 42% market share, far ahead of China (6%) and Israel (4%) (Figure 17). For many countries, buying American weapons is tied to security guarantees. Export volume is vital for smaller markets like Sweden and Israel, but less so for the U.S., which risks losing market share if allies prioritize quantity over quality or question U.S. security commitments. The "America First Arms Transfer Strategy" may further impact U.S. foreign military sales. These issues can be further compounded by high sustainment costs, delivery delays, and/or the "resequencing" of deliveries.

(vi) Government-Industry Link: This measures how effectively a nation's government can direct its defense sector to meet strategic objectives, especially as commercial technologies like AI and quantum computing become critical. China and Israel excel through military-civil fusion and dual-use mandates, ensuring reserve capacity for defense. The U.S. still operates largely on a buyer-vendor model, as seen in recent disputes with tech firms like Anthropic.

Asymmetric warfare

The war in Ukraine, and, to a degree, Iran, has become a real-time case study in modern conflict, revealing two intertwined lessons for defense industrial strategy.

First, the power of mass and attrition is clear. Drones and other cost-effective, attritable systems have shown how persistence and saturation can impose outsized costs on more expensive enemy assets. These systems aren't just substitutes; they sustain operational tempo, amplify sensing, and force adversaries to waste costly interceptors and reveal valuable positions. The battlefield now rewards rapid regeneration and asymmetric tactics, where affordable hardware can shift the balance.

Second—and arguably more critical—is the transformative impact of technology and software²⁶. Ukraine's forces have leveraged real-time data from distributed sensors, drones, satellites, and commercial sources to create a unified operational picture. Software compresses the kill chain, moving from slow, hierarchical targeting to fast, distributed loops powered by data integration. This shift has dramatically reduced time-to-target, enabling units to coordinate and survive through speed.

Command and control is now pivotal. Without a robust JADC2 infrastructure, even the most advanced platforms lose effectiveness. Stealth aircraft, long-range missiles, and ISR assets only deliver decisive results when connected to resilient data networks that enable coordinated, multi-domain action. Hardware is still essential, but software increasingly determines whether it delivers a decisive effect.

This reality underpins the emerging High-Low strategy: scalable, attritable systems (like UAVs) complement high-end assets (like F-35s), with their synergy driven by command, control, and data. The lesson from Ukraine is clear—modernization isn't just about hardware, but about integrating software, data, and adaptable manufacturing at speed and scale.

26McNamara, Whitney M., Peter Modigliani, and Tate Nurkin. Commission on Software-Defined Warfare: Final Report. Atlantic Council, Scowcroft Center for Strategy and Security, 2025

Box 3: Space-enabled services and JADC2

The war in Ukraine has shown that software-driven command and control, powered by space-enabled services, can be as decisive as any weapon platform²⁷. By compressing the kill chain—from sensing to decision to action—Ukraine leveraged space-based ISR and commercial imagery for timely warnings and sustained battlefield visibility. Intelligence from Western partners and U.S. providers like Maxar and Planet Labs proved critical, echoing the JADC2 vision: connect sensors into a unified network, process data (including with AI), and rapidly recommend the best response, as future decisions may need to happen in seconds.

The conflict also revealed the battlefield as a cycle of disruption and adaptation. On day one, a Russian cyberattack on Viasat knocked out satellite communications, forcing Ukraine to improvise. Starlink quickly became a vital SATCOM enabler, supporting tactical communications, command-and-control, drone operations, and fire support—demonstrating how resilient satellite internet can sustain coordination and tempo under attack. In JADC2 terms, this is the backbone: without robust communications, data can't flow fast enough to drive effective action.

Precision effects in Ukraine depended on space-provided PNT, but electronic warfare (EW) quickly challenged this advantage. GNSS signals were jammed and spoofed, sparking a constant contest over navigation and targeting. Ukraine, the U.S., and manufacturers adapted systems and tactics in real time, while Russia countered—proving that in an EW-rich environment, precision is a moving target, maintained through continual iteration. This underscores why JADC2 prioritizes rapid, software-enabled decision-making: when navigation, communications, and sensing are contested, the side that integrates data and adapts fastest keeps the kill chain intact.

The Imperatives: Integration, Scale, and Innovation

The U.S. defense-industrial enterprise now stands at a decisive inflection point. Success in the era ahead will be determined by its ability to deliver on three tightly interlocking imperatives: integrating the defense and commercial industrial bases, unlocking genuine mass-scale production, and unleashing innovation across the entire ecosystem. Together, these are the enabling conditions for a force capable of meeting the relentless operational and technological pressures revealed in Ukraine, exposed by China's fusion model, and only latently present in the U.S. system today.

Integration: A single, responsive IB

In our view, the key foundational imperative is the deep integration of the defense and commercial industrial bases. This would allow resilience, rapid fielding, and long-term technological superiority, which in turn would mean moving beyond episodic, transactional procurement toward structural alignment in three critical mechanisms:

²⁷Radin, Andrew, et al. Lessons from the War in Ukraine for Space: Challenges and Opportunities for Future Conflicts. RAND Corporation, 2025

Leveraging Commercial Manufacturing Platforms: Integration means defense products must be designed from inception for manufacturability on widely available commercial lines, rather than bespoke supplier sets dictated by narrow military-unique requirements. The goal is not simply outsourcing, but genuine modularity—allowing programs like the Barracuda missile (see Appendix: IV. “High and Low” Strategy - The Barracuda Missiles) to be produced across multiple U.S. manufacturing sites, harnessing the scale, process control, and cost discipline of the commercial domain.

Shared Supplier Development and Open Standards: Prime contractors must overhaul relationships with tier-2 and tier-3 suppliers, investing in their capabilities, workforce, and technical tools, potentially mirroring the ASML open innovation model (see **Box 4: ASML’s Open Innovation Model** below). Establishing common interface standards would reduce barriers for new entrants and enable the commercial supply chain to flow into defense, fostering competition and reducing single points of failure.

Cross-Sector Feedback Loops: Commercial velocity in software, AI, and automation must be continuously imported into the defense design and production cycle—not on a lag, but as the default setting. Feedback from operational use, testing, and adversary adaptation must be rapidly recycled into requirements, integrating lessons learned at every level. This is the mechanism by which the U.S. can maintain its advantage as the tempo of warfare, software, and data integration accelerates.

Box 4: ASML’s Open Innovation Model

Founded in 1984 and headquartered in the Netherlands, ASML is the world’s dominant manufacturer of semiconductor lithography equipment. Its product portfolio spans extreme ultraviolet (EUV) and deep ultraviolet (DUV) lithography systems, computational lithography, and metrology and inspection systems, with operations across more than 60 locations on three continents. Despite being a monopoly, holding 100% market share in EUV lithography and approximately 80–89% across all lithography tool categories, ASML has remained at the technological frontier by building a business model around collaborative partnerships, shared risk-reward arrangements, and deep integration with customers, suppliers, and research institutions—an “open innovation” strategy (ASML, 2026)—supplemented by targeted vertical integration.

The most prominent example of ASML’s “open innovation” strategy is its Customer Co-Investment Program for Innovation, established in 2012, through which its three largest customers — Intel, TSMC, and Samsung — acquired equity stakes in ASML while simultaneously committing R&D funding and gaining board-level governance participation (ASML, 2012). A comparable logic underpins ASML’s relationship with Carl Zeiss AG, its longest-standing institutional partner and the supplier of the precision optics at the heart of every ASML lithography system.

In 2016, ASML formalized the relationship by acquiring a 24.9% minority stake in Carl Zeiss SMT, while committing €220 million in R&D support and approximately €540 million in capital expenditure and supply chain investments over six years (ASML, 2016). Most recently, ASML extended this co-investment model beyond its traditional hardware ecosystem with a €1.3 billion investment in Mistral AI’s Series C funding round, acquiring approximately a 11% stake and a seat on the company’s Strategic Committee (ASML, 2025). ASML extends this open innovation logic further through a comprehensive network of public-private

partnerships that embed the company within academic and governmental R&D efforts (e.g., Project Beethoven and IMEC partnership).

Complementing this ecosystem-level moat, ASML has systematically integrated critical capabilities through targeted acquisitions. Its first major acquisition, Silicon Valley Group (SVG) in 2001, created the world's leading lithography equipment provider and added photoresist track and thermal product lines to ASML's portfolio (ASML, 2000). The 2013 acquisition of Cymer internalized EUV light source development, eliminating a key external dependency and accelerating ASML's EUV commercialization roadmap (ASML, 2013). In 2016, the acquisition of Hermes Microvision Inc. (HMI) in Taiwan integrated pattern verification and inspection capabilities, enabling ASML to offer holistic lithography solutions combining exposure, metrology, and inspection in a unified product architecture (ASML, 2016).

Finally, the 2020 acquisition of the Berliner Glas Group secured internal manufacturing of specialized optical glass components essential to EUV systems, converting a strategically critical external supply relationship into a controlled internal function (ASML, 2020). Beyond these targeted vertical integrations, ASML maintains a highly diversified supplier base, with 80% of its materials sourced from 5,100 suppliers (ASML Annual Report, 2025). Each acquisition thus follows a disciplined logic of internalizing only those capabilities that represent genuine bottlenecks to ASML's roadmap, while the expansive supplier network ensures the company retains the resilience and flexibility that full vertical integration would undermine.

Scale: Modularity and scalability

Integration loses much of its purpose without scale. The system must deliver not just exquisite prototypes but repeatable, robust, mass production of attritable systems while also being capable of delivering on high-end systems. The Ukraine War has demonstrated that victory depends on the ability to field thousands of effectors and continuously regenerate lost capability, an industrial challenge for which the traditional U.S. defense base is currently unprepared.

The key to scale lies in two linked reforms:

Modularity and Interoperability: Systems, particularly in the “low” (attritable) tier, must be modular in design, interchangeable in subsystems, and as close as feasible to open technical standards. This approach allows for rapid reconfiguration and taps the full breadth of U.S. manufacturing capacity. Modularity, even at the exquisite tier, matters: next-generation platforms must be upgradable—hardware and software alike—enabling continuous integration rather than serial block upgrades. The standardization and process repetition delivered by modularity is a precondition, as well, for unlocking the full mass gains of advanced production methods such as robotic manufacturing ([Revitalizing U.S. shipbuilding: Automation, autonomous, partnerships, and shipping](#)).

Scalable Pathways for New Entrants: Innovation from commercial-origin firms and new defense manufacturers remains at risk unless structural bottlenecks (requirements gold-plating, slow contracting, supply chain credentials) are cleared. Reducing barriers allows these actors to move beyond prototypes to true volume production. Prime contractors must foster—rather than inhibit—these transitions, treating the broader

industrial base as a networked resource, not just vendors of convenience.

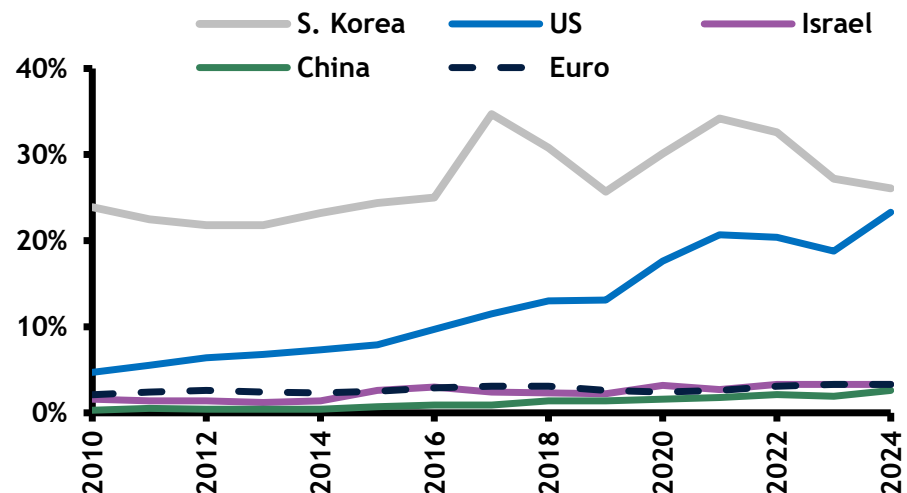
Scale in this regard offers both a solution to the attrition needs of the modern battlefield and is also the precondition for fielding enough (and adaptable) exquisite platforms, and modularity plays a large role in enabling this.

Innovation: Software, Hardware, Ecosystem

Finally, as we have argued, the pace of innovation—not just in technology, but in process, integration, and organizational adaptation—will be a decisive differentiator. The U.S. is in a privileged position in terms of software talent, digital infrastructure, and advanced hardware development; the challenge is structural, not technical (Figure 19).

Figure 19: AI firms' share of economy-wide capex

% (Publicly-listed AI firms' capex divided by total publicly-listed firms' capex)



Source: BIS, J.P. Morgan.

Software-Driven Transformation: Software is now a warfighting element in its own right—command and control, sensor fusion, targeting, and electronic warfare all depend on its rapid evolution. Modular software design, open mission systems, and digital engineering must become core practices, with new acquisition models to match.

Iterative Hardware Development: Attributable systems are valuable not only for their volume, but for the rate at which they can be adapted and improved. The model must shift from monolithic, decades-long programs to iterative spirals, driven by robust data from operational use, threat developments, and technological advances. Connecting hardware and software cycles enables rapid learning—again, a virtue best delivered by a deeply integrated, open defense ecosystem.

Dynamic Exquisite Systems: The “high” tier of U.S. capability—the stealth aircraft, advanced munitions, and space systems—remains essential, but its value erodes if it cannot adapt quickly. These platforms must adopt commercial partnership models, modular upgrades, and institutional openness to outside innovation. Modeling this integration on successful commercial exemplars offers the best prospects for agility and enduring advantage.

New names challenge incumbent primes

Some of the drawbacks of industry concentration, outlined in the preceding sections, are already being addressed by an emerging defense tech ecosystem that diverges from the incumbent defense industrial base and offers a potentially complementary production model. Drawing largely on commercial technologies and productive capacities, this ecosystem seeks to fuse the leading edge of software with modern manufacturing processes to deliver a new array of military hardware and software—one designed to meet the autonomy, adaptability, and, potentially, the scale and attrition demands of contemporary warfare. In doing so, it embodies many of the principles highlighted earlier: modularity, iterative production, rapid adaptability, deep software integration, and scalability.

Crucially, it also offers a chance to disrupt the concentration, sluggishness, and limited competition that have come to characterize the exquisite-platform-based infrastructure, fostering a broader and more networked defense base through reproducibility, design simplicity, and lower unit costs—qualities exemplified by programs like the Barracuda missile family (see Appendix: IV. “High and Low” Strategy - The Barracuda Missiles).

Some big players are already emerging within this space—Anduril, Shield AI, Saronic, Forterra, and others have secured billions in capital and increasingly sizeable Pentagon contracts, positioning themselves as central partners in deploying autonomous systems at scale. This rapid ascent introduces a non-negligible risk on two fronts: first, that the new and traditional ecosystems grow along parallel tracks rather than becoming integrated in ways that foster competition and adaptability across the board; and second, that today’s primes are simply replaced by a new generation of neo-primes, reproducing the very concentration dynamics this ecosystem promises to disrupt. The new ecosystem must therefore be carefully nurtured, through institutional reform, open standards, and procurement models that preserve openness and competition, if it is to deliver on its promise rather than recreate the limitations of the system it aims to complement.

Defense procurement is shifting from platform-centric programs toward software-defined architectures, spawning a new class of cheaper, scalable hardware with less human involvement. Attritable platforms (low-cost, expendable/reusable uncrewed systems) are integrating with software-led kill chains like Palantir’s Maven Smart System and Anduril’s Lattice. Leading players — Shield AI (air), Saronic (maritime), Forterra (ground), and Anduril (all domains) — have secured billions in capital and hundreds of millions in contracts, positioning themselves as essential Pentagon partners for deploying thousands of autonomous systems. These systems operate autonomously in execution with human-in-the-loop oversight. The DoD’s FY2026 budget allocated \$13.4 billion for autonomous systems: \$9.4B for UAVs, \$1.7B for surface water, \$734M for underwater, \$210M for ground vehicles, and \$1.2B for enabling capabilities.

1. Air Domain and the CCA Program. The air domain is the most critical and developed arena for hardware-software integration. UAV pure-play companies raised over \$8B in equity (2016–2025), with 90%+ concentrated in U.S. startups (Table 1).

Shield AI has built its position around Hivemind, a proprietary autonomy stack enabling aircraft to operate without GPS, communications, or a remote pilot. The V-BAT is a vertical takeoff and landing drone for ISR and strike with 12+ hours

endurance, built for contested EW environments. V-BAT has executed hundreds of targeting operations in Ukraine under GPS and communications jamming. Shield AI is competing for up to \$800M in Navy ISR task orders deploying V-BAT from ships.

Table 1: Unmanned air vehicles - U.S. companies

US\$ million, except per patent number

Company	Year founded	Total Raised	Total Patent Documents	Revenue	Last Known Valuation	Last Financing
Anduril Industries	2017	6,870	376	4,300	30,000	30
Shield AI	2015	3,160	1	540	12,700	1,500
Skydio	2014	966	925	180	4,400	110

Source: PitchBook, J.P. Morgan

Skydio's X10D flies autonomously without GPS using onboard navigation cameras, delivering persistent ISR with high-resolution visual and thermal sensors day and night. Skydio secured the largest single-manufacturer small UAS procurement in U.S. Army history (\$52M+ for 2,500+ drones). **Anduril's** UAS portfolio spans Altius loitering munitions, Fury jet-powered autonomous aircraft, and Roadrunner reusable interceptor. Anduril signed a 10-year enterprise contract with the U.S. Army valued at up to \$20B, consolidating its full stack (software, hardware, services) into a single framework covering drones, counter-drone interceptors, sensor networks, and Lattice C2.

The CCA program — manned fighters supported by swarms of lower-cost autonomous aircraft for ISR, EW, and strike — is transitioning from prototype to production with \$2.37B requested in FY27. Shield AI was selected as the mission autonomy provider, integrating Hivemind into Anduril's Fury — best-in-class autonomy software paired with purpose-built attritable hardware from a separate vendor to deliver a unified combat system.

2. Maritime Domain. The second most developed arena. Autonomous surface and subsurface vessels are maturing from experimental concepts to production-ready platforms, driven by the Navy's need for distributed, persistent presence at lower cost. Pure-play companies raised ~\$2B (2016–2025), 90%+ U.S.-based.

Saronic builds autonomous surface vessels ranging from the 24-foot Corsair to the 180-foot Marauder. The Navy awarded Saronic a \$392M Corsair production contract, moving from prototype-to-production in under 12 months. Saronic is investing \$300M expanding its Franklin, Louisiana shipyard for high-rate production. **Saildrone** builds long-endurance USVs deployed by the Navy since 2021. The 52-meter Spectre targets ASW, surveillance, and strike (built by Fincantieri Marine Group). Lockheed Martin invested \$50M in Saildrone (October 2025) to integrate defense payloads into multi-mission combat systems (Table 2).

Anduril's Ghost Shark is an AI-powered autonomous submarine for the Royal Australian Navy, backed by A\$1.7B (~\$1.1B) in Australian government investment with delivery by 2028, extending the attritable, software-defined model into undersea warfare. In March 2026, the Navy announced the MUSV Family of Systems program, opening the marketplace to vendors. The MUSVs will carry at least two 40-foot container-style payloads for missions ranging from long-range strike to anti-ballistic missile defense, targeting 5–10 production vessels by FY2027.

Table 2: Unmanned water vehicles - U.S. companies

US\$ million, except for number of patents

Company	Year founded	Total Raised	Total Patent Documents	Revenue	Last Known Valuation	Last Financing
Saronic	2022	2,595	29	13	9,250	1,750
Saildrone	2012	246	21	48	588	60
HavocAI	2024	197	--	--	784	100

Source: Pitchbook, J.P. Morgan

3. Ground Domain. This is the least developed arena — just \$210M of the \$13.4B FY2026 autonomy budget — though it is gaining traction. The Army’s RCV and Marine Corps’ ROGUE-Fires are driving demand for autonomous ground platforms in unstructured, GPS-denied environments.

Forterra’s AutoDrive is the most widely deployed autonomous driving system in defense, fielded across 12+ vehicle platforms including combat zones, and integrated into ROGUE-Fires. The MESA is a modular autonomous vehicle (Polaris Ranger XD 1500-based) for last-mile military capabilities.

Kodiak Robotics adapted its commercial trucking autonomy stack (Kodiak Driver) for defense, completing a \$30M three-year Army RCV contract. In early 2026, the Marines awarded Kodiak a ROGUE-Fires carrier vehicle contract. Kodiak partnered with General Dynamics Land Systems on the Leonidas AGV, combining Kodiak’s autonomy with Epirus’ HPM for mobile counter-drone operations. Both Forterra (JLTVs) and Kodiak (carrier vehicle) serve ROGUE-Fires in complementary roles, mirroring CCA’s multi-vendor model (Table 3).

Table 3: Unmanned ground vehicles

US\$ million, except for number of patents

Company	Year founded	Total Raised	Total Patent Documents	Revenue	Last Known Valuation	Last Financing
Forterra	2002	556	222	30	1,000	238
Kodiak Robotics	2018	448	199	4	1,377	100

Source: Pitchbook, J.P. Morgan

4. Counter-UAS, Air Defense, and Directed Energy. Low-cost drone swarms (\$20K–\$50K each) create a fundamental cost asymmetry against \$4M Patriot interceptors. Iran has used mass drone attacks to overwhelm Israeli and Gulf state defenses. The Pentagon requested \$20.6B for counter-UAS in FY2027.

Anduril’s Lattice fuses sensor data and coordinates layered defenses from EW to autonomous interceptors. The Marines awarded Anduril \$642M for Installation Counter-UAS; its \$20B Army contract also covers counter-drone systems. **Epirus’** Leonidas HPM system neutralizes drones and swarms without kinetic munitions at near-zero cost-per-shot. The Army awarded \$43.5M (July 2025) for next-gen systems expected to more than double range and increase power 30% over the four units delivered in 2024 (Table 4).

AeroVironment’s 2025 **BlueHalo** acquisition created a cross-domain defense platform. Its counter-UAS portfolio includes the Titan 4 (autonomous RF detect-and-defeat), Locust directed-energy laser, and Freedom Eagle-1 (FE-1) selected as the Army’s Next-Generation Counter-UAS Missile. The Army awarded \$95.9M for FE-1, targeting

\$150K–\$200K unit cost to close the cost gap against Shahed-type drones costing tens of thousands to produce.

Table 4: Counter-UAS, air defense, and directed energy weapons - U.S. companies

US\$ million, except for number of patents

Company	Year founded	Total Raised	Total Patent Documents	Revenue	Last Known Valuation	Last Financing
Blue Halo*	2020	1,372	140	1,000	3,485	852
Epirus	2018	540	117		1,100	250

Source: Pitchbook, J.P. Morgan

5. Lessons from the Replicator. Launched in August 2023 to field thousands of expendable autonomous systems by August 2025, Replicator reportedly produced only hundreds. The corrective course: building manufacturing capacity (Anduril’s Arsenal-1 in Columbus, Ohio; Saronic’s Louisiana shipyard), investing in mature autonomy software (Hivemind, Kodiak Driver, AutoDrive, Lattice), and using enterprise contracts and open marketplaces to incentivize private capital and rapid production. The lesson: deploying at scale likely requires not just urgency and innovation, but an ecosystem that is only now being built.

Institutional Challenges

Institutional reform is the gatekeeper for industrial transformation. Recognizing the urgent need for structural change, the U.S. government has enacted executive orders, memoranda, and provisions in the FY26 **National Defense Authorization Act** (NDAA) to accelerate innovation, streamline requirements, integrate commercial solutions, and align software acquisition with iterative development and rapid deployment.

Against the backdrop of multi-domain conflict, compressed decision cycles, and a battlefield dominated by software, data, and resilient networks, the NDAA’s commercial-first approach aims to bridge DoW demand with the agile, scalable capabilities of the broader U.S. industrial sector.

Commercial First

The DoW has traditionally defaulted to defense contractors and bespoke, defense-unique solutions, even when commercial products are available, because requirements, compliance, and security processes tend to favor established primes and discourage commercial vendors from participating. The FY26 NDAA aims to reverse this trend. It enforces a commercial-first approach by requiring thorough research and formal documentation before opting out of commercial products and services, and it reduces the compliance burden that has historically deterred commercial suppliers by limiting defense-unique clauses in contracts and restricting how far these obligations can be pushed down the supply chain.

The NDAA also expands the use of Commercial Solutions Openings and follow-on production authorities, making commercial adoption feasible at scale rather than just for prototypes. In parallel, the DoD increasingly uses Other Transaction Agreements (OTAs) to engage nontraditional vendors and move quickly outside the rigid FAR/DFARS framework.

To lower barriers that favor incumbents, the NDAA raises thresholds for cost-accounting and pricing regimes, focuses compliance requirements on the largest programs, and updates evaluation practices to better include new entrants—such as recognizing commercial past performance and relying more on demonstrations and testing. It also provides broad exemptions for nontraditional defense contractors from traditional cost-accounting and business-system requirements, aiming to widen the competitive field and allow more commercial companies to participate without disrupting their business models.

For software, the NDAA moves acquisition toward continuous delivery by establishing software-appropriate test and evaluation pathways and enabling commercial-style consumption and subscription models, acknowledging that modern software advantage comes from ongoing iteration rather than one-time deployment.

IP rights and budget cycles

Procedural reform alone doesn't guarantee operational success. Take MOSA (Modular Open Systems Approach): it's DoW's statutory framework for building modular, upgradeable systems with open standards to reduce vendor lock-in and enable faster refresh cycles. But MOSA's effectiveness hinges on clear, workable approaches to data and intellectual property (IP) rights—specifically, how data must be delivered and what can remain proprietary. Without a robust IP strategy, contracts stall, negotiations drag on, and vendors either price in risk or opt out entirely. These boundaries remain hotly contested, often undermining MOSA's intended benefits.

Even with improved acquisition authorities, the resourcing system, governed by Planning, Programming, Budgeting, and Execution (PPBE), still dictates what gets funded, when, and how flexibly resources can be redirected as technology and threats evolve. PPBE's annual cycles and rigid appropriation categories weren't designed for rapid iteration, continuous software delivery, or quick adoption of commercial tech. While execution flexibility may improve slightly, the fundamental mismatch persists: commercial innovation moves fast, but defense budgeting is locked in early and constrained by “colors of money,” line-item controls, and reprogramming hurdles. Without a budget model that supports the iteration modern warfare demands, this could turn procurement reform into little more than a permission slip.

Ultimately, implementation is where reform succeeds or stalls, and where resistance is strongest. Institutional inertia, compliance culture, and legacy incentives slow progress. Structural barriers like budget categories and documentation requirements, plus the tendency to revert to familiar processes, further impede change. Political concerns about oversight can also limit flexibility. Without real follow-through that transforms day-to-day budgeting and execution, the commercial-first, software-centric agenda risks stalling not in law, but in the practical mechanics of funding, contracting, and acquisition incentives.

Future-proofing U.S. defense

The strategic necessity facing the U.S. is clear: only through deep integration of commercial and defense sectors can scale and innovation be restored, and only through institutional reform can technological agility be matched to operational need. The lessons drawn from peer and partner ecosystems all point in the same direction. China's

military-civil fusion, Israel's tight coupling between defense firms and the civilian high-tech ecosystem, Sweden's reliance on trusted allied supply networks, and Korea's public-private partnership model each demonstrate, in their own way, that defense effectiveness in the modern era flows from the porousness of the boundary between commercial and military industrial bases, not from their separation.

The U.S. retains a decisive advantage in the inputs that matter most for the wars of the future: software talent, digital infrastructure, advanced semiconductor design, AI capabilities, and a deep commercial manufacturing base. The challenge is not the absence of these assets, but the structural friction that prevents them from flowing into the defense ecosystem at the speed and scale that modern warfare demands. Closing that gap is less a technical problem than an institutional and organizational one.

The era ahead will be defined not by any single tier of capability, but by the synergy among them. Attributable systems have proven their worth in Ukraine, Iran, and the Middle East, where mass, persistence, and cost asymmetry have repeatedly shifted operational balances against more expensive adversary platforms. Yet the exquisite tier—stealth aircraft, advanced munitions, nuclear submarines, space systems—remains essential, and will continue to anchor deterrence and high-end warfighting. What unites the two is the connective tissue: command, control, communications, and the software-driven kill chains that allow sensors, decision-makers, and shooters to operate as a single distributed system.

The U.S. is already seeing the rise of a promising new defense tech ecosystem—Anduril, Shield AI, Saronic, Forterra, Epirus, and others—that can complement its globally-leading exquisite tier with attributable, software-defined hardware produced on commercial-style manufacturing lines. But this ecosystem will not deliver on its promise automatically. It must be nurtured and integrated with the incumbent base rather than left to grow on a parallel track, and shaped by procurement and policy choices that preserve openness and competition rather than allowing today's primes to be replaced by a new generation of neo-primes.

In scale and speed, the U.S. has fallen behind, and the costs of inaction are tangible: thinned tier-2 and tier-3 supplier ranks, lengthy acquisition timelines that field obsolete electronics, fragile supply chains exposed to Chinese chokepoints, and a replicator initiative that produces hundreds rather than thousands of systems. If these gaps are not closed, the U.S. risks both failing to deter adversaries, who increasingly read American industrial constraints as a window of opportunity, and losing future conflicts in which the side that iterates, regenerates, and adapts fastest holds the decisive edge.

Institutional change is building momentum: the FY26 NDAA's commercial-first provisions, expanded use of Other Transaction Agreements, MOSA, and software-appropriate acquisition pathways all point in the right direction. But operational follow-through is far from assured. Procurement reform can be undone by PPBE rigidity, IP and data-rights disputes, compliance culture, and the gravitational pull of legacy incentives. Both the institutional and productive challenges remain tall, and they are deeply intertwined, resolving the former is a precondition for delivering on the latter. The next few years will determine whether America's defense industrial base evolves into the integrated, scalable, software-driven enterprise the moment requires, or whether reform stalls in the practical mechanics of funding, contracting, and execution. The window for getting this right is narrowing.

Appendix

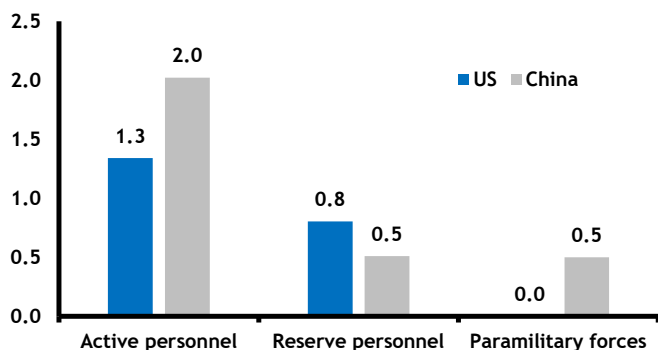
I. U.S. and Chinese Military Capabilities

Comparing U.S. and Chinese military capabilities exemplifies how each state's resources translate into usable power. While both are positioned as the world's leading military competitors, their force structures reflect different strategic problems shaped over decades: the U.S. has focused on projecting, sustaining, and commanding forces across distant theaters, whereas China has prioritized regional advantage, internal security, and the ability to contest access near its periphery.

China's manpower advantage is most visible in active personnel (Figure 20). That scale supports large standing formations and persistent domestic and regional presence, consistent with a continental power managing extensive borders and internal stability requirements. The U.S. offsets a smaller active force with a larger reserve component, which aligns with a model that can expand capacity through mobilization and rotational deployments. China also dominates in paramilitary forces, which indicates a structurally distinct emphasis on internal security and regime stability while also providing additional military depth.

Figure 20: Military personnel

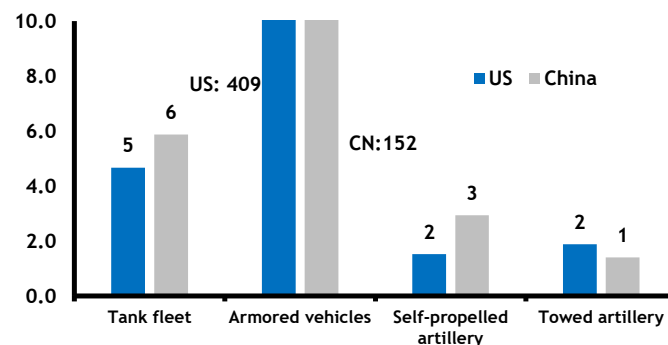
Million forces



Source: IISS, J.P. Morgan

Figure 21: Land assets

Thousand



Source: IISS, J.P. Morgan

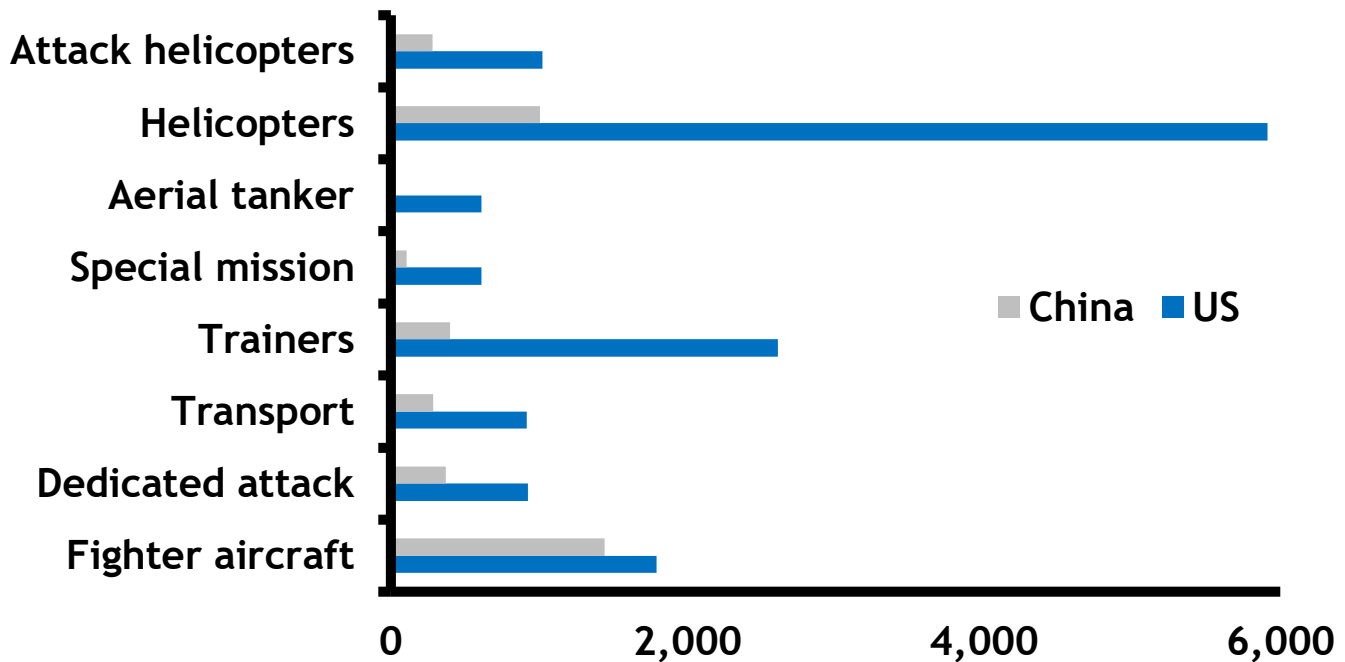
Land capabilities present a mixed picture consistent with different strategic priorities (Figure 21). China leads in heavy armor (e.g., tanks) and self-propelled artillery, which supports large-scale ground combat and border defense scenarios where massed armored and artillery formations remain central. The U.S., on the other hand, holds a significant advantage in overall armored vehicle inventory. This suggests depth in mechanized mobility, protected transport, and combined-arms maneuver capability that supports expeditionary operations, force protection, and sustained overseas deployments.

The U.S. holds its clearest quantitative dominance in air power, with nearly a 4x advantage vs. China in terms of aircraft fleet size (Figure 22). This gap is especially meaningful because it is concentrated in categories that enable global operations, not only combat platforms (aerial tankers, transport aircraft, special mission aircraft). The fighter gap is comparatively narrower than the overall fleet disparity (1,791 US vs.

1,443 Chinese), suggesting that China has built meaningful combat mass in core airframes even while remaining far behind in the logistics and support architecture required for sustained operations far from home. The U.S. also leads heavily in helicopters, assets that enable maneuver, close support, and distributed operations.

Figure 22: Air assets

Units



Source: IISS, J.P. Morgan

U.S. naval dominance is most pronounced in aircraft and helicopter carriers, reinforcing the ability to generate air power at sea across multiple theaters, as well as in major surface combatants and undersea forces that typically underpin sustained blue-water operations (Table 5). Collectively, these categories align with a fleet designed to control sea lanes, deter globally, and sustain operations far from home ports. China, by contrast, outnumbers the U.S. in several categories associated with regional maritime control and coastal operations (frigates, corvettes, patrol vessels, and mine/countermine warfare ships). This mix supports a strategy centered on near-seas dominance, maritime security, and challenging adversary access. China's ability to grow and refresh these categories quickly is plausibly linked to industrial scale, particularly shipbuilding capacity and modernization velocity, which can translate into faster hull production and fleet expansion.

Table 5: Naval Assets

Units

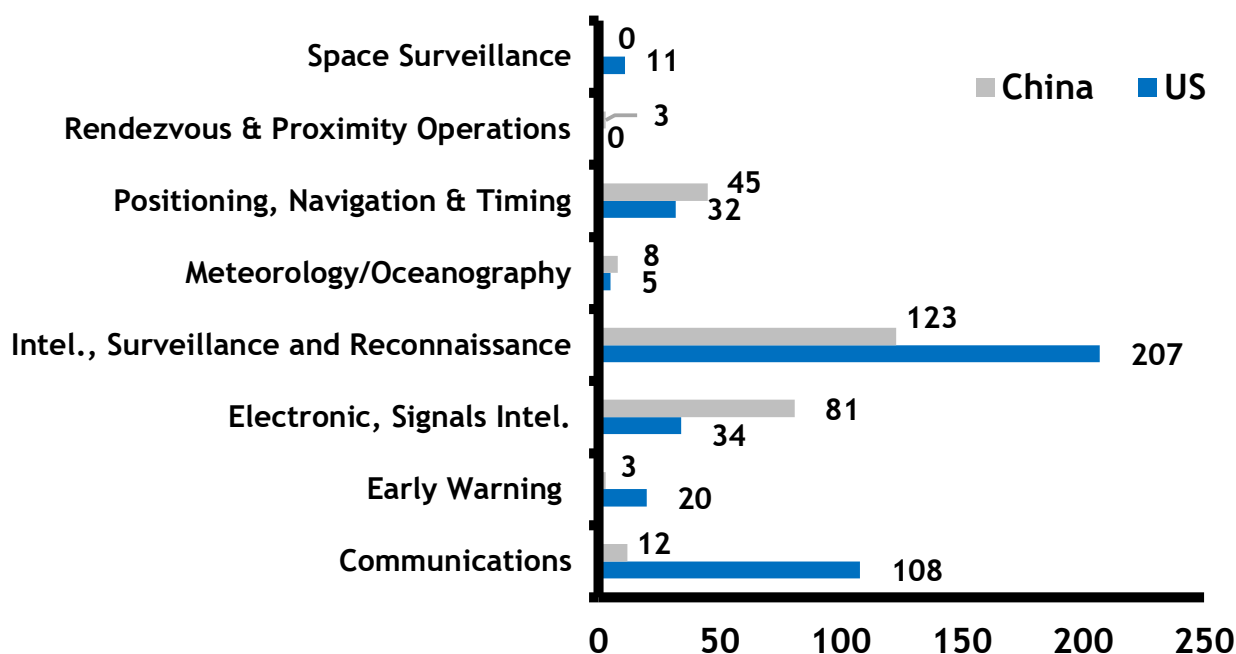
	U.S.	China
Aircraft carriers	11	3
Helo carriers	9	4
Submarines	66	61
Destroyers	83	53
Frigates	0	46
Corvettes	27	50
Patrol vessels	0	150
Mine/countermine warfare ships	4	36

Source: IISS, J.P. Morgan

The fleet age profile reinforces this industrial advantage in stark terms. China has launched at least 168 naval vessels since 2010, compared to just 63 for the U.S. over the same period. The contrast deepens when looking at older segments of the fleet: the U.S. still operates 101 warships launched between 1985 and 1999, including vessels dating back to the early 1970s, while China has virtually no active ships launched before 1995. In practical terms, China's fleet is predominantly 21st-century, while over two thirds of the U.S. naval fleet is at least 15 years old.

Figure 23: Space military assets (satellites)

Units



Source: IISS, J.P. Morgan

Space assets reflect a similar divide (Figure 23). The U.S. operates the larger active military satellite constellation, with pronounced advantages in communications, early warning, and Intelligence, Surveillance, and Reconnaissance (ISR). These categories

underpin the real-time command, precision targeting, and situational awareness required to coordinate forces across distant theaters. China, by contrast, leads in electronic and signals intelligence satellites, positioning, navigation and timing (PNT), and is one of only two operating rendezvous and proximity operations satellites (used to support space situational awareness, surveillance, and inspection of other space objects), a mix oriented toward monitoring adversary activity, reducing dependence on U.S.-controlled systems like GPS, and developing the ability to interfere with hostile space assets in case of conflict.

Between 2016 and 2025, the U.S. launched roughly 13,000 satellites compared to about 1,400 for China. While the vast majority of U.S. satellites launched are commercial (well over three quarters), the fact that launching companies are domiciled in the US gives it a degree of strategic leverage. While these assets are not government-owned, the fact that Starlink is a U.S.-domiciled enterprise with existing defense contracts means the U.S. retains a degree of strategic leverage (in a conflict scenario, it could theoretically repurpose or prioritize commercial communication capacity for military use). On the other hand, China leads in civil satellite launches, which are government-owned satellites for non-military purposes, suggesting a dual-use investment strategy where nominally civilian assets may serve security functions.

The U.S. force structure signals a military designed to project and sustain combat power overseas, with air and naval enablers. China's structure emphasizes manpower depth combined with a land and naval fleet mix well-suited to regional control and security. Fleet age data further underscores China's industrial and modernization momentum. Its navy is not only numerically strong in regional categories but substantially newer on average than the U.S. fleet. The net implication is not that one side "wins" across all categories, but that each dominates where its strategy demands. The U.S. dominates in global reach and sustained operations overseas, and China in regional security depth and border protection.

II. Drones & Robotics in Modern Warfare

Drones have emerged as a cornerstone of modern warfare, propelling the global defense drone market into a phase of extraordinary expansion, with annual sales surpassing \$50 billion and a growth rate of 20% CAGR ([JPMorgan Research](#)). This surge is largely driven by the demands of high-intensity attrition warfare, as witnessed in the Russia-Ukraine conflict and the ongoing escalation between the U.S., Israel, and Iran in Operation Fury. The Russia-Ukraine war has firmly established drones as a principal tool of "strategic attrition." By early 2026, reports indicate that Ukrainian drone strikes have disabled nearly 17% of Russia's oil refining capacity, demonstrating how high-volume, cost-effective systems can circumvent traditional air defenses and inflict substantial economic and military damage ([Link](#))²⁸. The deployment of drones in the current U.S.-Iran conflict further underscores a "race to the bottom" in U.S./Israeli munitions stockpiles. These drone deployments have illuminated the critical importance of cost asymmetry, innovation, and, above all, supply chain and manufacturing self-sufficiency. For example, the economic imbalance is stark against the U.S.: intercepting a \$20,000 Shahed-type drone from Iran with a \$2 million Patriot missile is

28.Russian oil refineries, terminals burn as Ukraine hits Putin's war economy, Reuters (2025)

unsustainable. But drone innovation is evolving from singular, monolithic platforms to adaptive, autonomous swarms: AI-integrated drones capable of real-time coordination without manual piloting to overwhelm advanced air defense systems.

While the U.S. possesses the innovative capability and an evolving policy framework favorable to responding rapidly to the evolving threat of drone warfare, concerns persist regarding manufacturing capacity and supply chain resilience. Critical components such as integrated circuits and batteries remain largely sourced from China, which continues to dominate global electronics production. For instance, in 2024, Skydio, one of the largest U.S. drone manufacturers, reported supply chain strains and rationing of batteries during assembly of their drones due to Chinese sanctions affecting U.S. battery availability.

Market Landscape

The deployment of drones and robots across commercial and military domains is reshaping the global landscape of defense innovation. While the U.S. defense industry has historically led the way in drone technology, exemplified by the MQ-9 Reaper drone which was first deployed in the Afghanistan war in 2007, the market has since evolved dramatically. Today, consumer and commercial platforms, which are often repurposed for military use, dominate the field, as vividly demonstrated during the Russia-Ukraine war and during the current conflict between U.S./Israeli versus Iran. During the course of the Ukrainian war, First Person View (FPV) drones, Quadcopters such as the DJI Mavic 3, and others built with commercially available components have been widely deployed by both Russia and Ukraine. Despite the U.S.'s early innovation advantage in drone technology, China has emerged as the preeminent producer of low-cost, dual-use drones, leveraging its formidable manufacturing capacity and supply chain.

In response to these shifts, the U.S. government has launched several initiatives to strengthen domestic drone manufacturing. The Drone Dominance Initiative, introduced in June 2025 with a \$1.1 billion investment, aims to bolster the U.S. drone industry. This and similar programs like the Replicator Initiative that aims to field swarms of affordable autonomous attritable drones across air, land, and sea to counter China's military modernization, have accelerated the development of platforms such as [SpektreWorks](#)' LUCAS drones, now available for military procurement.

Among other robotics segments, Humanoid Robots have emerged as a technological breakthrough, and are rapidly moving from R&D to real-world deployment. Although initial applications are geared towards research, manufacturing, logistics, retail, and consumer segments, the military applications of these robots are eminent. In China, for example, government procurement and policy initiatives are driving order momentum ([Link](#)). Chinese companies such as Unitree, UBtech, and XPeng lead the segment in cost and scale, while U.S. firms such as Boston Dynamics and Tesla lead in setting technical benchmarks. As global competition and the integration of AI into humanoid robots accelerates, both the U.S. and China are beginning to deploy humanoid robots on the battlefield. For example, in the U.S., Foundation Robotics Labs is partnering with the U.S. DoW to develop their Phantom MK-1 robots for the battlefield ²⁹. Similar military applications of humanoid robots have been reported in China.

²⁹<https://time.com/article/2026/03/09/ai-robots-soldiers-war/>

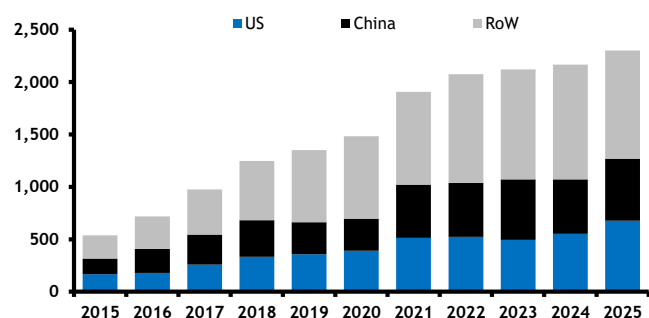
However, the opacity of the industry complicates analysis of the market landscape. Many leading drone & robotics companies, including China's SZ DJI Technologies, are privately held and do not disclose manufacturing capacity, making it difficult to obtain a comprehensive view of the market. To gain more insight into the market, we therefore conducted a market analysis using PitchBook's database and insights from CSET ([Source](#)), examining the landscape, major players, private financing, and other market trends. The PitchBook dataset encompasses Robotics and Drones players including manufacturers, assemblers, software providers, and key component suppliers.

A maturing market

Our analysis identified over 2,000 companies—public and private—operating globally in the robotics and drone sector by 2025, with the U.S. and China accounting for 29% and 26% of these firms, respectively (Figure 24). The industry is overwhelmingly dominated by private enterprises, which comprise 97% of the global total. Private investment reached a record \$67 billion in 2025, reflecting a robust 13% CAGR since 2015 (Figure 25). The U.S. led global investment, contributing 63% of total funding, while China accounted for 15%. Notable private companies from both countries are detailed in Sections IV and V.

Figure 24: Drone companies by country

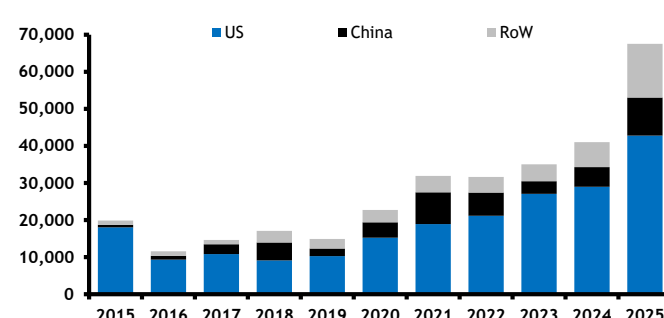
Number of companies



Source: Pitchbook, J.P. Morgan

Figure 25: Private capital invested by country

US\$ million

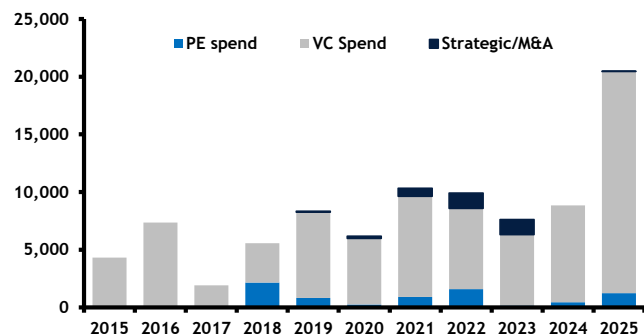


Source: Pitchbook, J.P. Morgan

Venture capital has been the primary engine of private investment. In 2025, U.S. venture capital investments totaled \$19 billion, representing 94% of all U.S. sector investments. This marked a pivotal year, with private investment more than doubling compared to 2021—the previous peak (Figure 26). Late-stage venture capital dominated, comprising 74% of total VC activity, underscoring the sector's technological maturity and its imminent transition to mass adoption (Figure 27).

Figure 26: Private capital in U.S. robotics and drones

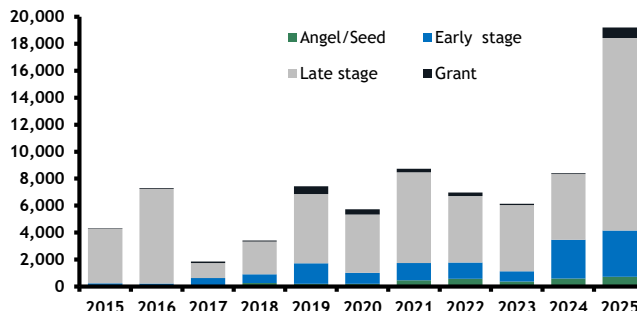
US\$ million



Source: Pitchbook, J.P. Morgan

Figure 27: U.S. VC funding of robotics and drones

US\$ million



Source: Pitchbook, J.P. Morgan

Capabilities and challenges of the U.S. industry

According to a CSET analysis ³⁰, most U.S. drone companies focus on small UAVs (Groups 1–3, as classified by the DoW), while only a handful of defense firms develop more complex military systems (Groups 4–5), as summarized in Table 6. Challenger Aerospace Systems (acquired by Resilient Energy in 2024), UAVOS, General Atomics, and Northrop Grumman are among the few producing military-grade drones. Yet, as the Ukraine conflict has shown, commercial drones (Groups 1–3) can be readily adapted for battlefield use (Figure 28).

Our analysis suggests that the U.S. possesses the innovative capacity to develop drones and robots for a wide range of military applications. However, persistent supply chain chokepoints and manufacturing capacity must be addressed. It remains unclear to what extent U.S. drone production relies on foreign components, as highlighted by Skydio’s 2024 battery supply challenges. Moreover, in scenarios akin to the Ukraine war—where up to 7,000 drones per day were deployed—the question of whether the U.S. drone industry can achieve self-sufficient, large-scale manufacturing of this magnitude during a prolonged war remains unresolved (Gen. Petraeus, Aspen Institute 2025). In our view, this can only be achieved by deploying the manufacturing capacity available outside of the traditional defense industrial base.

Table 6: U.S. marketed UAVs by type

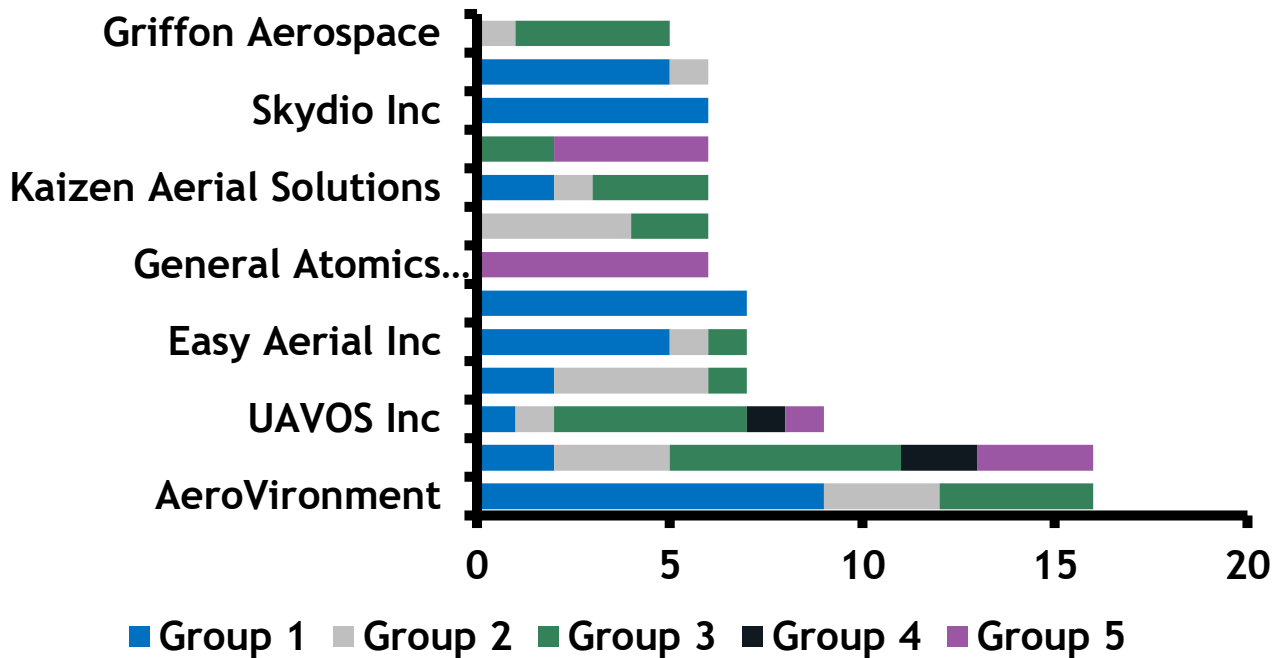
UAV group	Max gross takeoff weight (lbs)	Normal operating altitude (ft)	Airspeed (knots)
Group 1	0 - 20	Less than 1,200	Less than 100
Group 2	21 - 55	Less than 3,500	Less than 250
Group 3	Less than 1,320	Less than 18,000	Less than 250
Group 4	More than 1,320	Less than 18,000	Any airspeed
Group 5	More than 1,321	More than 18,000	Any airspeed

Source: CSET

³⁰Kyle Miller, Sam Bresnick, Jacob Feldgoise, and Christian Schoeberl, (2025), “The US aerial drone market”, CSET

Figure 28: U.S. marketed UAVs by group and company

Number of UAVs



Source: CSET

III. The F-35 - A Signature Program

Historical Background

The F-35 Lightning II (Figure 29), born from the Joint Strike Fighter (JSF) initiative, is among the DoD's most ambitious acquisition programs and a centerpiece of U.S. airpower modernization. Conceived as a fifth-generation aircraft family, it was designed to deliver stealthy, multirole fighters that could replace a wide range of aging platforms across the U.S. services and close allies, while using commonality such as shared parts, software, training, and support, to reduce long-term operating costs.

This strategic logic took shape in the post-Cold War period, when the U.S. and its partners confronted a dual challenge: large, aging tactical fighter inventories and tightening budgets that made separate, bespoke replacement programs for each service increasingly untenable. Rather than building three wholly distinct aircraft for the Air Force, Navy, and Marine Corps, DoD planners pursued a single core design adaptable into multiple variants. The program's central wager was that scale and standardization, reinforced by broad international participation, could make an advanced stealth platform affordable to procure in meaningful numbers and sustain over decades.

Contractors and Industrial Base

The JSF competition began in 1993 and culminated in 2001 when Lockheed Martin's X-35 was selected over Boeing's X-32, formally setting the course for the F-35. The aircraft's first flight in 2006 marked the transition from concept demonstration to a

prolonged phase of systems integration, flight testing, and iterative maturation of mission systems and variant-specific performance. In broad terms, it took roughly 13 years from program start to maiden flight, and more than two decades from inception to initial operational service in 2015. While the lengthy timeline is partly a reflection of both the technical ambition and the complexity of delivering three variants from one baseline architecture, it has also come to epitomize the typical long U.S. weapon system development timelines, which in today's evolving warfare are seen as a meaningful challenge to U.S. defense readiness.

Figure 29: The F-35, refueling mid-air



Source: US Air Force

Economically, the program's value chain is anchored by Lockheed Martin as prime contractor and lead integrator. Pratt & Whitney (RTX) holds the propulsion franchise through the F135 engine. Among the major tier-one partners, Northrop Grumman contributes major structures including the center fuselage and participates in mission systems and radar-related elements, while BAE Systems is similarly positioned with major responsibilities often described as including the rear fuselage and key electronics. This structure creates a concentrated value-capture model: although thousands of suppliers participate, a small set of primes and tier-one partners control the subsystems that drive differentiation, integration risk, and much of the lifecycle sustainment economics. Lockheed Martin has cited a global supply chain of 1,900+ companies. While the U.S. remains the principal manufacturer and largest customer, the program has expanded into a broad coalition of buyers—roughly 19 customer countries—spanning partners and foreign military sales customers. More than 1,300+ aircraft have been delivered globally, reflecting the platform's scale and the durability of its multinational demand base.

Key capabilities and technology

The F-35 is engineered around survivability, flexibility, and information advantage. Its foundation is a low observable (stealth) design, intended to reduce detection and targeting by advanced sensors and integrated air defenses. It is built as a true multirole aircraft, spanning air-to-air missions, precision strike, ISR (intelligence, surveillance, and reconnaissance) contributions through onboard sensing, and elements of electronic

warfare. As a single-engine supersonic fighter, its speed is estimated at Mach 1.6, balancing speed and maneuverability with a design optimized for signatures, sensors, and integrated weapons employment.

What most differentiates the F-35 is the architecture of its mission systems. Its signature capability is sensor fusion, the integration of radar, infrared, and electronic warfare inputs into a single, coherent operational picture intended to reduce pilot workload and accelerate decisions. This is reinforced by 360-degree situational awareness, enabled by distributed sensors (often described as a Distributed Aperture System), allowing detection and tracking across all aspects of the battlefield. The aircraft is also designed for network-centric operations, sharing and receiving near real-time information with other platforms and forces, which extends its role beyond delivering weapons to acting as an airborne node in a joint and coalition kill chain. Finally, internal weapons bays are central to the F-35's survivability concept, enabling weapons carriage while preserving low observability compared to external stores.

Program Cost

The F-35's performance ambitions have been accompanied by persistent scrutiny over cost growth and schedule pressure. Total lifetime program costs are estimated to exceed \$2.1 trillion through an expected horizon extending to 2088, with a per-aircraft cost cited at approximately \$90 million. The most consequential driver, however, is sustainment – an estimated 75% of total costs accrue from operations, maintenance, and repairs, underscoring that the program's long-run economics are dominated by readiness, supply chain performance, and depot capacity rather than procurement alone. For example, program costs increased from about \$1.1 trillion in 2018 to \$1.58 trillion 2023—a 44% increase—which was attributed to the compounding burden of a complex global supply chain and the logistics required to maintain availability over time.

Challenges faced in a future war with a peer adversary

Despite its advanced technology and central role in U.S. force planning, the F-35 has clear liabilities in a peer conflict defined by A2/AD threats, long distances, and heavy electronic/cyber contestation—with direct operational and industrial-base implications. First, range and geography are structural constraints. An Indo-Pacific fight would likely force U.S. and allied aircraft to operate from more distant bases due to missile threats to forward airfields, carriers, and tankers. This increases the F-35's dependence on aerial refueling and distributed basing; if tankers are degraded or pushed back, combat reach, persistence, and sortie rates fall. Second, the F-35's edge depends on connectivity. Its core advantage comes from sensor fusion and data sharing, but a peer adversary can attack the enabling networks through EW, cyber, and anti-satellite actions. In a degraded communications environment, the aircraft remains capable, but its differentiating advantage shrinks, weakening the payoff case for its cost.

Third, sustainment and availability can be decisive. High maintenance demands, spare-parts constraints, and depot bottlenecks can translate into a gap between fleet size “on paper” and sorties (recovering, refueling, rearming and launching) generated in practice, especially under wartime stress and disrupted logistics. The platform's complex, specialized supply chain and reliance on a sophisticated maintenance ecosystem amplify this risk. Fourth, cost limits scale and attrition tolerance. The F-35 delivers high capability per aircraft, but its expense and complexity constrain procurement volumes and make loss replacement and rapid regeneration more difficult than with simpler

platforms—creating a potential quality-versus-quantity mismatch in a prolonged conflict. Finally, counter-stealth improvements (e.g., networked sensors and IR detection) may not negate stealth, but they can erode its margin and raise detection risk in dense air-defense environments. Overall, the F-35 is best viewed as one node in a larger ecosystem, and its peer-conflict performance is contingent on tankers and basing access, resilient networks, robust sustainment/depot capacity, and sufficient force mass to sustain operations.

IV. “High and Low” Strategy - The Barracuda Missiles

Background

In 2024, the U.S. Air Force, in collaboration with the Defense Innovation Unit, launched the Enterprise Test Vehicle (ETV) Program—a pioneering initiative aimed at identifying and prototyping commercial off-the-shelf (COTS) components to alleviate supply chain bottlenecks and advance dual-use technology solutions. The program’s central objective is to establish a modular, cost-effective air vehicle architecture that will serve as the foundation for large-scale production of a new generation of missiles.

Participating vendors are integrating COTS components wherever feasible, not only to streamline supply chains and reduce costs, but also to embrace contemporary design-for-manufacturing principles. This approach ensures that air vehicles are purpose-built for their missions, avoids unnecessary complexity and the use of costly materials, and enables rapid, on-demand production—capabilities unattainable with more exquisite, bespoke counterparts such as Tomahawk missiles.

The Defense Innovation Unit selected four companies, including Anduril Industries, to develop prototype solutions for flight demonstration. Remarkably, within just seven months, Anduril refined its Barracuda-500 missile design and successfully conducted a flight test (Figure 30). This achievement underscores how Barracuda’s simple, modular architecture accelerates the development, testing, and deployment of disruptive capabilities for warfighters. The Barracuda missile family—including the -100, -250, and -500 variants—embodies air-breathing, autonomous vehicles engineered for high-rate production and modular payload integration, setting a new standard for agility and scalability in defense manufacturing. This makes the Barracuda substantially cheaper (~10x cheaper) than the Tomahawk missile.

Figure 30: The Barracuda 500



Barracuda-500

Source: Anduril

Strategically, the Barracuda is not a replacement for the Tomahawk missile system because it doesn't have the range and payload capacity as the Tomahawk. But it is used to supplement more exquisite systems with cheap munitions used in large scale attrition attacks such as for drone warfare. Barracuda is purpose-built to bring mass to the fight. Designed from the ground-up for hyper-scale manufacturing, its production rates can be doubled on-call to meet short-term surges in demand. Barracuda is designed to be simple to manufacture, both in terms of tooling and talent. This is another example of the implementation of the DoD's high-low acquisition strategy, in which a limited number of high-cost, advanced (exquisite) platforms is paired with a large number of low-cost, disposable assets to enhance military effectiveness while controlling costs.³¹

Technical specifications for the Barracuda-500

- Propulsion: 150 pound thrust Turbo jet engine
- Speed: 190 –500 knots
- Range: over 500nautical miles
- Payload: 100+ pounds of kinetic and non-kinetic payloads
- Mounted on fighter aircrafts such as F-16 and also ground launched

Production efficiency

- Ten or fewer tools to assemble – The Barracuda-500 also has a minimized number of parts to accelerate assembly ensuring flexibility and simplicity that allows it to be produced by commercial automotive and consumer electronics workforce.
- Supply chain resiliency built into design – The Barracuda missiles leverages commercially and widely-available components across six common sub-systems.
- Rapid hardware integration & testing – The Barracuda has a modular open system architecture that enables rapid integration and testing with the latest advancements in commercial sub-component technology to optimize mission systems against evolving threats
- Software-defined upgradability – The Barracuda is a software-defined, hardware-enabled platform with rapid software updates for mission-specific needs and autonomous behaviors

³¹ <https://www.anduril.com/news/anduril-unveils-barracuda>

Procurement

In April 2026, the DoD launched a request for information (RFI) for a Family of Affordable Mass Missiles – Beyond Adversary’s Reach (FAMM-BAR). The FAAM program (including the FAMM-P and FAMM-L programs) was established as the program of record by building off the Enterprise Test Vehicle (ETV) program which included Anduril and other companies such as Zone 5. Though details on the program are not readily available, the Air Force is requesting \$55 million in discretionary and \$300 million in mandatory funds in FY27 to buy 1,000 copies of FAMM weapons, with expectations to increase this number in the coming years according to DoD budget estimates.

V. Main global public defense companies*

Table 7: Global public defense companies

US\$ million, 2024 (except where noted)

Company (c)	Company Description	Country	Arms revenues	Total revenues	Arms revenues as a % of total	Market Cap/ Valuation (billion)
Lockheed Martin Corp.	Lockheed Martin is the world's largest defense company and is the manufacturer of the F-35 jet. Aeronautics is Lockheed's largest segment. Other segments include rotary and mission systems - mainly the Sikorsky helicopter; missiles and fire control, and space systems.	United States	64,650	71,040	91.0%	118
RTX	RTX is an A&D manufacturer formed from the merger of United Technologies and Raytheon, with roughly equal exposure across 3 segments: Collins Aerospace - a diversified aerospace supplier; Pratt & Whitney - aircraft engine manufacturer; and Raytheon - providing missile systems, sensors, and communications to the military.	United States	43,600	80,740	54.0%	240
Northrop Grumman Corp.	Northrop Grumman is a diversified defense contractor providing aeronautics, defense, and space systems. The aerospace segment produces aircrafts. Defense systems makes missile ammunition systems. Mission systems creates radar, navigation, and communication systems. Space systems produces satellites, space structures, and rocket motors.	United States	37,850	41,030	92.2%	78
BAE Systems	BAE Systems is a British global defense, security, and aerospace company and the largest defense contractor in Europe. The company has five operating segments: electronics systems, the cyber, security, and intelligence segment; platforms and services; the air segment; and the maritime segment.	United Kingdom	33,790	35,400	95.5%	77
General Dynamics Corp.	General Dynamics is a defense contractor and business jet manufacturer. The firm's segments are aerospace, marine, combat systems, and technologies. The aerospace segment manufactures Gulfstream business jets. Combat systems produces land-based combat vehicles such as the M1 Abrams tank.	United States	33,630	47,720	70.5%	93
Boeing	Boeing is a major aerospace and defense firm operating in three segments- commercial airplanes; defense, space, and security; and global services.	United States	30,550	66,520	45.9%	188
Rostec	Rostec is a developer, manufacturer and exporter of high-tech industrial products. The company operates in areas that include automotive manufacturing, aircraft manufacturing, engine manufacturing, metallurgy, construction, optics, composite and other modern structural materials, etc.	Russia	27,120	38,890	69.7%	-
AVIC	A state-owned publicly traded aerospace & defense company with 100 subsidiaries. It operates in 4 divisions: AVIC Aircraft, China Aviation Industry General Aircraft, Avicopter, China National Aero-technology import & export.	China	20,320	81,290	25.0%	-
CETC	A state-owned company that provides communication equipment, computers, electronic equipment, IT infrastructure, networks, software development, and research services for civilian and military applications	China	18,920	55,230	34.3%	-

L3Harris Technologies	L3Harris Technologies has, through a series of acquisitions, established franchises producing unmanned aerial vehicles, sensors, avionics, space-based systems, missiles, and solid rocket motors.	United States	16,210	21,330	76.0%	56
NORINCO	A state-owned defense corporation that manufactures commercial and military products	China	13,970	61,580	22.7%	
Leonardo	Leonardo is one of the largest European defense firms, with 30% of its shares owned by the Italian government. The group's divisions include helicopters; defense, electronics, and security systems; and aeronautics.	Italy	13,830	19,210	72.0%	35
Airbus	Airbus is a global aerospace and defense firm that designs, develops, and manufactures commercial and military jets and helicopters, as well as space launch vehicles and satellites. The company operates in three divisions: commercial, defense and space, and helicopters.	Trans-European	13,370	74,890	17.9%	163
CSSC	CSSC is a state-owned shipbuilding conglomerate and consists of various shipyards, equipment manufacturers, research institutes, and shipbuilding-related companies that build both civilian and military ships.	China	12,330	49,630	24.8%	-
Thales	Thales is a French aerospace and defense industrial and one of Europe's largest defense contractors. The company has three segments: defense and security; aerospace; and digital identity and security.	France	11,800	22,260	53.0%	54
HII	Huntington Ingalls Industries is the largest independent military shipbuilder in the U.S., spun out from Northrop Grumman in 2011. It operates 3 segments: Ingalls produces non-nuclear-powered ships; Newport News produces nuclear-powered ships; and shares production of destroyers and nuclear submarines with Bath Iron Works and Electric Boat shipyards, respectively.	United States	10,280	11,540	89.1%	13
CASC	It is state-owned and a main contractor of the Chinese space program. It subsidiaries design, develop, and manufacture a range of spacecraft launch vehicles and ground equipment.	China	10,230	34,100	30.0%	-
Leidos	Leidos Holdings Inc is a technology, engineering, and science company that provides services and solutions in the defense, intelligence, civil, and health management, both domestically and internationally.	United States	9,370	16,660	56.2%	16
Amentum	Amentum Holdings Inc is an engineering and technology solutions provider serving U.S. and allied government agencies, as well as other customers across energy and environmental, intelligence, space, defense, civilian and commercial markets.	United States	8,330	13,860	60.1%	6
Rheinmetall	Rheinmetall focus is on the security technology and mobility segments. It operates in five divisions: vehicle systems, weapons and ammunition, electronic solutions, sensors and actuators, and materials and trade.	Germany	8,240	10,550	78.1%	65
Hanwha Group	It is a Korean business conglomerate with diversified holdings in explosives, energy, materials, aerospace, etc.	South Korea	7,970	64,100	12.4%	9
Booz Allen Hamilton	Booz Allen Hamilton Holding Corp provides technology solutions in areas such as artificial intelligence, cybersecurity, and related fields. The company serves U.S. federal government agencies, commercial clients, and select international customers.	United States	7,810	11,980	65.2%	9
Rolls-Royce	Rolls-Royce operates 3 core business segments: civil aerospace, power systems, and defense. The civil aerospace segment builds aircraft engines. Power systems provides power solutions to defense, agriculture, marine, and power generation; while the defense business provides military, ground vehicle and naval propulsion solutions.	United Kingdom	7,200	22,810	31.6%	139
CACI International	CACI International Inc is an information solutions and services provider, offering information solutions and services to its customers. The company's primary customers are agencies and departments of the U.S. government, which account for the vast majority of the firm's revenue.	United States	6,510	8,630	75.4%	11
Elbit Systems	Elbit Systems Ltd is a technology company involved in producing a portfolio of systems and products for aircraft, land, and naval applications. The company's products are used for defense, homeland security and commercial flight capabilities. The company operates in five segments: Aerospace; C4I and Cyber; ISTAR and EW; Land; and ESA.	Israel	6,280	6,830	91.9%	37
AECC	It is a state-owned aerospace and defense company specializing in the research, development, and manufacturing of aircraft engines, gas turbines, and related aerospace materials.	China	6,260	..	..	

Honeywell International	Honeywell is a global multi-industry company which operates through 4 segments: aerospace technologies, industrial automation, energy and sustainability solutions, and building automation.	United States	6,090	38,500	0.16	139
Saab	Saab specializing in advanced aerospace, weapons systems, and radar technology	Sweden	5,550	6,030	0.92	30
Safran	Safran designs, develops and manufactures both commercial and military aircraft engines; launch vehicle, spacecraft and missile propulsion systems; as well as various other aerospace and military equipment and devices.	France	5,320	29,550	0.18	137
MBDA	MBDA specializes in the design, development and manufacturing of missiles and related systems. It is a joint venture owned by Airbus, BAE Systems, and Leonardo.	Trans-European	5,260	5,310	0.99	-
Israel Aerospace Industries	Israel Aerospace Industries is a developer and manufacturer of aerial and astronautic systems for both military and civilian usage. IAI is completely state-owned by the government of Israel.	Israel	5,190	6,110	0.85	-
Mitsubishi Heavy Industries	MHI's products include aerospace and automotive components, air conditioners, elevators, forklift trucks, hydraulic equipment, printing machines, missiles, tanks,[3] power systems, ships, aircraft, railway systems, and space launch vehicles.	Japan	5,030	33,190	0.15	92
GE Aerospace	GE Aerospace is the global leader in designing, manufacturing, and servicing commercial aircraft turbine engines, along with partner Safran in their CFM joint venture.	United States	4,890	38,700	0.13	314
Rafael	Rafael is state-owned and is known for developing combat-proven systems like the iron dome.	Israel	4,700	4,800	0.98	-
KBR	KBR is a global provider of technology, integrated engineering, procurement, and construction delivery, and operations and maintenance services. The business is organized into two segments: mission technology and sustainable technology solutions.	United States	4,680	7,740	0.60	4
EDGE Group	Manufacturer of missile equipment and weapons intended for the defense sector. The company offers air platforms, land systems, naval systems, electronic warfare and other related products for cyber defense, secure communications, sub-systems, training & services and precision engineering companies.	United Arab Emirates	4,660	4,900	0.95	-
Naval Group	Manufacturer and designer of submarines, surface ships and marine defense systems. The company's operations include manufacturing, developing and designing of naval infrastructures, weapons, nuclear and diesel submarines, corvettes, frigate and command ships, drones and maritime surveillance systems.	France	4,660	4,710	0.99	-
CSGC	It is a state-owned manufacturer of automobiles, motorcycles, firearms, vehicle components, and optical-electronic products and other special products.	China	4,580	44,680	0.10	-
Babcock International Group	Babcock is a British aerospace, defence and nuclear engineering services company.	United Kingdom	4,570	6,170	0.74	7
Dassault Aviation Group	It is controlled by the French Dassault Group, and designs, builds, and maintains military fighter jets and is one of the largest manufacturers of business jets.	France	4,290	6,740	0.64	26
United Shipbuilding Corp.	It manufactures, supplies, maintains, upgrades, repairs, and recycles military and civilian shipping and facilities and offers commercial vessels and offshore equipment.	Russia	4,110	4,960	0.83	-
KNDS	Manufacturer of land defense systems intended for modern land forces in Germany. The company's products include weapons and turrets, armored vehicles, artillery systems, infantry fighting vehicles, light wheeled vehicles, and support systems.	Trans-European	4,110	4,110	1.00	-
Science Applications International Corp.	SAIC provides technical, engineering, and enterprise IT services mainly to the U.S. government. Specifically, it offers end-to-end solutions spanning the design, development, integration, deployment, management, operations, sustainment, and security of IT infrastructure.	United States	3,890	7,480	0.52	4
Hindustan Aeronautics	The company is engaged in the design, manufacturing, and development of products like aircraft, helicopters, systems and accessories, power plants, avionics, materials, and aerospace structures	India	3,810	4,010	0.95	34
V2X	V2X Inc provides services to the U.S. government. It operates as one segment and offers facility and logistics services and information technology mission support and engineering and digital integration services.	United States	3,760	4,320	0.87	2
Czechoslovak Group	CSG NV is a defence group that produces medium, large- and small-calibre ammunition.	Czechia	3,630	4,340	0.84	18

ASELSAN	Turkey's largest defense electronics company, delivering advanced technology solutions in defense and security.	Türkiye	3,470	3,660	0.95	43
Textron	Textron is a conglomerate that designs, manufactures, and services a range of specialty aircraft including small jets, propeller-driven airplanes, helicopters, and tilt-rotor aircraft.	United States	3,290	13,700	0.24	16
TransDigm Group	It manufactures and services a broad set of specialized parts for commercial and military aircraft. The operates in 3 segments: power and control, airframes, and a small non-aviation segment, which serves mostly off-road vehicles and mining equipment.	United States	3,180	7,940	0.40	67
NCSIST	A state-owned corporation, formerly part of the Ministry of National Defense's Armaments Bureau, which is active in the development, manufacturing, support, and sustainment of various weapons systems and dual use technologies.	Taiwan	3,110	3,190	0.97	-
PGZ	PGZ is a holding company established by the Polish government in 2013 to unite Polish state-owned defence industry companies.	Poland	3,040	3,380	0.90	-
JSC Ukrainian Defense Industry	It is the largest arms manufacturer in Ukraine	Ukraine	3,010	3,010	1.00	-
Fincantieri	It is one of the world's shipbuilding groups, operating across the defense, commercial, and specialized maritime sectors. The company designs and builds a wide range of vessels, including cruise ships, naval combatants, submarines, and specialized ships.	Italy	2,990	8,790	0.34	5
Palantir	Palantir is an artificial intelligence, analytics, and automated decision-making company that leverages data to drive efficiency across its clients' organizations. The firm serves commercial and government clients via its Foundry and Gotham platforms, respectively.	United States	NA	2,865	NA	329
Parker-Hannifin Corp.	The firm produces aerospace engines, agricultural and construction machinery, freight and passenger vehicles, and industrial automation equipment.	United States	2,740	19,850	0.14	110
Kawasaki Heavy Industries	It is a conglomerate with operations in aerospace, marine, and vehicles. The company operates through 6 segments: The Aerospace Systems segment; The Energy Solutions & Marine segment; The Power Sports & Engine segment; The Precision Machinery and Robots segment; The Vehicle segment manufactures.	Japan	2,650	14,060	0.19	17
Bechtel Corp.	Provider of engineering, construction, and project management services to industrial and government clients.	United States	2,640	23,000	0.11	-
ST Engineering	It is government-linked commercial and defense engineering group. Its key businesses include aircraft maintenance, repair and overhaul services, in which it is the world's largest independent third-party provider.	Singapore	2,620	8,440	0.31	26
Bharat Electronics	It is an aerospace and defense company predominantly owned by the Indian government. It manufactures a wide range of specialized electronics and equipment for defense and civilian use.	India	2,470	2,750	0.90	33
Serco Group	It is a British multinational defence, health, space, justice, migration, customer services, and transport company.	United Kingdom	2,420	6,760	0.36	4
LIG Nex1	It produces a wide range of advanced electronic systems, including missiles, underwater weapon systems, radars, electronic warfare, avionics, tactical communication systems, fire control systems, naval combat systems, and electro-optics.	South Korea	2,400	2,400	1.00	15
ThyssenKrupp	It is a German industrial engineering and steel production company. Its products range from machines and industrial services to high-speed trains, elevators, and shipbuilding.	Germany	2,290	37,900	0.06	8
Hensoldt	It specializes in defense and security electronics, operating across two main segments: sensors and optronics.	Germany	2,240	2,420	0.93	10
Parsons Corp.	It is a provider of technology-driven solutions in the defense, intelligence, and critical infrastructure markets.	United States	2,220	6,750	0.33	5
Fujitsu	The company operates in three segments. The Hardware Solution segment; Service Solution, and Ubiquitous Solutions	Japan	2,190	23,440	0.09	37
Oshkosh Corp.	It is a leading maker of access equipment, specialty vehicles, and military trucks.	United States	2,160	10,730	0.20	8
Turkish Aerospace Industries	Manufacturer of aerospace systems which offers design, development, modernization, manufacturing, integration, and life cycle support of integrated systems, from fixed and rotary wing air platforms to UAVs and satellites.	Türkiye	2,160	2,700	0.80	-
Diehl	Manufacturer of metal, control, defense, aero system, and metering products intended for customers in Germany and internationally.	Germany	2,110	5,080	0.42	-

BWX Technologies	It is a specialty manufacturer and service provider of nuclear components, operating in two segments: Government Operations and Commercial Operations.	United States	2,070	2,700	0.77	19
Teledyne Technologies	The firm operates in four segments: Digital Imaging, Instrumentation, Aerospace and Defense Electronics, and Engineered Systems.	United States	2,050	5,670	0.36	29
Korea Aerospace Industries	The company builds military aircraft for national defense and civil aircraft parts. The company operates through 4 segments: Airframe, Fixed Wing, Rotary Wing, and Others.	South Korea	2,010	2,640	0.76	11
QinetiQ	QinetiQ Group PLC is a company operating in the aerospace, defense, and security sectors. The services and products offered are air; C4ISR; Cyber; Maritime.	United Kingdom	1,950	2,470	0.79	3
Sierra Nevada Corp.	The company offers aircraft integration and modification solutions, electromagnetic spectrum systems, joint integration architecture in the fields of aviation, national security space, electronic warfare, command and control, mission systems and inline cybersecurity.	United States	1,940	2,000	0.97	-
Baykar	Operator of an aerospace company intended to produce robust, high-performance unmanned systems.	Türkiye	1,900	2,000	0.95	-
General Atomics	Developer of a high-technology system designed for nuclear fuel cycle to electromagnetic system.	United States	1,890	..	..	-
Eaton	Eaton's portfolio can broadly be divided into two parts: its electrical and industrial businesses. Its electrical portfolio sells components within data centers, utilities, and commercial and residential buildings, while its industrial business sells components within commercial and passenger vehicles and aircraft.	United States	1,870	24,880	0.08	163
Mitsubishi Electric Corp.	It is a leading diversified industrial manufacturer, specializing in factory automation systems, power and energy systems, infrastructure solutions, automotive equipment, electronic devices, and home appliances.	Japan	1,850	36,460	0.05	84
Curtiss-Wright Corp.	It delivers engineered products and services to commercial, defence, power generation, and other industrial markets.	United States	1,790	3,120	0.57	27
Kongsberg Gruppen	It is an international technology company focused on defense, aerospace, and advanced sensing and surveillance solutions.	Norway	1,780	4,550	0.39	28
Hyundai Rotem	It manufactures and sells railway vehicles, defense systems, and plants and machinery in South Korea and internationally. The Defense sector supplies key ground equipment such as K2 tanks and wheeled armored vehicles.	South Korea	1,730	3,210	0.54	16
CNNC	It is a state-owned enterprise which oversees all aspects of China's civilian and military nuclear programs.	China	1,670	38,750	0.04	-10
Amphenol Corp.	Amphenol is a global supplier of connectors, sensors, and interconnect systems.	United States	1,670	15,220	0.11	150
United Launch Alliance	Provider of spacecraft launch services.	United States	1,540	..	..	-
NEC Corp.	NEC Corp is engaged in IT services, networks, social infrastructure, and solutions. The company operates through 3 segments: IT Services, Social Infrastructure, and Others.	Japan	1,540	22,600	0.07	35
CAE	Manufacturer of simulation technologies, modelling technologies and training services to airlines, aircraft manufacturers, and defence customers.	Canada	1,460	3,440	0.42	8
Moog	Manufactures precision motion and fluid control systems for aerospace, defense, and industrial markets. Its four segments are Space and Defense, Military Aircraft, Commercial Aircraft, and Industrial	United States	1,400	3,610	0.39	10
Roketsan	Manufacturer and defense contractor to establish Turkey's industrial base on missile systems.	Türkiye	1,390	1,390	1.00	-
Navantia	Operator of a military shipbuilding construction company. The company offers civil and military naval construction, ship repair, and manufacturing of turbines, engines, weapons systems, and more.	Spain	1,270	1,650	0.77	-
HEICO Corp.	An aerospace and defense supplier that focuses on creating replacement parts for commercial aircraft and components for defense products.	United States	1,250	3,860	0.32	34
Melrose Industries	Melrose Industries is a UK-based company that manufactures engines and structural products for all major OEMs across civil and defense markets.	United Kingdom	1,240	4,430	0.28	-
Mazagon Dock Shipbuilders	The company is engaged in building and repairing ships, submarines, various types of vessels, and related engineering products for its customers.	India	1,230	1,370	0.90	11

ViaSat	Provides bandwidth technologies and services in three segments: communication services and defense and advanced technologies.	United States	1,220	4,520	0.27	10
MKE	The corporation mainly produces equipment for the Turkish Armed Forces, such as the ammunition for small arms and heavy weapons, artillery systems, aerial bombs, mines, explosives, and rockets.	Türkiye	1,210	1,210	1.00	-
Howmet Aerospace	Offers engineered solutions for the aerospace and transportation industries, including castings for jet engines and industrial gas turbines; seamless rolled rings for jet engines.	United States	1,160	7,430	0.16	109
Keysight Technologies	Keysight specializes in the communications market, but also supplies into the government, automotive, industrial, and semiconductor manufacturing markets. Keysight's solutions include testing tools, analytical software, and services.	United States	1,150	4,980	0.23	63
DEFEND ID	A state-owned defense company launched in 2022 to strengthen Indonesia's defense capabilities.	Indonesia	1,140	1,640	0.70	-
TTM Technologies	Manufactures technology products, including mission systems, RF components, RF/microwave assemblies, and advanced interconnect products such as PCBs and substrates.	United States	1,130	2,440	0.46	17
Mitre Corp.	Not-for-Profit that operates federally funded research and development centers to tackle complex challenges in national security, aviation, etc	United States	1,120	.	.	-
The Aerospace Corp.	The company operates an independent, systems engineering and integration federally funded research and development centre for the space enterprise.	United States	1,100	1,400	0.79	-
Ultra Electronics Group	Provider of electronic and software technology services intended for the defense, aerospace, security, cyber, transportation and energy markets.	United Kingdom	1,070	1,240	0.86	-

Source: SIPRI, Bloomberg Finance L.P., J.P. Morgan. *Includes state-owned companies.

VI. Main U.S. private defense companies

Table 8: U.S. Private defense companies

Company	Company description
Anduril Industries	Developer of autonomous defense technology featuring UAVs, surveillance towers, sensor networks, and command software with real-time data integration, providing military and government agencies with scalable situational awareness and border security solutions.
Signia Aerospace	Manufacturer of aerospace thermal management systems and mission-specific equipment components, serving both aerospace and defense sectors with precision-engineered solutions.
Shield AI	Defense technology company deploying AI-powered autonomous systems capable of executing unscripted missions in denied environments, from room clearance to air defense penetration, without direct operator input.
World View	Stratospheric exploration company offering on-demand, high-resolution sensing via patent-protected balloon technology, providing persistent coverage and daily imaging as an affordable alternative to satellites and drones.
Saronic	Manufacturer of fully integrated unmanned surface vehicles combining naval hardware, software, and AI into a scalable platform for maritime security, domain awareness, and defense surveillance operations.
Sierra Space	Developer of reusable spaceplanes, hypersonic technologies, propulsion systems, and satellites for commercial, DoD, and national security clients, enabling reliable human space exploration and national security missions.
Stoke Space	Developer of fully reusable, low-cost rockets designed for daily orbital launches, delivering airline-like frequency and reliability to enable sustainable, on-demand satellite deployment and space mission access.
Dataminr	AI-powered real-time event detection platform analyzing public data signals across multiple languages to identify emerging threats and high-impact events, enabling corporate security, public sector, and news organizations to accelerate crisis response.
BlueHalo	Advanced engineering manufacturer for the national security community, offering autonomous systems, laser weapons, directed energy, cyber, and space superiority solutions across the full R&D-to-deployment lifecycle.
True Anomaly	Developer of AI-integrated autonomous spacecraft systems supporting orbital situational awareness, secure space access, and long-term sustainability for defense agencies, allied governments, and commercial space operators.
CHAOS	Advanced distributed detection and effects technology company integrating time-synchronized coherent networks, multi-domain sensor systems, and software-defined controls to deliver situational awareness and coordinated response for defense clients.
Skydio	Developer of AI-powered autonomous drones using computer vision to navigate and avoid obstacles in real time, enabling safe and intelligent aerial operations for defense and commercial users.
SandboxAQ	Developer of quantum-AI technology offering cryptographic management, AQ-enabled navigation, and large quantitative models, enabling quantum-resistant cybersecurity, drug discovery, and GPS-denied navigation across defense and commercial sectors.
Astranis	Manufacturer of small high-orbit satellites for protected defense communications, resilient GPS, and commercial connectivity, serving oil and gas, telecom, sovereign, and defense clients with robust satellite infrastructure.

PennAero	Manufacturer of precision aerospace and semiconductor components including fasteners, gears, manifolds, and latches made from metals and composites for airframe, engine, and semiconductor equipment applications.
MAG Aerospace	Provider of manned and unmanned ISR services including aerial fire, air charter, ISR operations and training, and C2 center maintenance, delivering real-time situational awareness for defense and government customers.
Mach Industries	Developer of VTOL strike aircraft, high-altitude gliders, satellite payloads, and counter-UAS interceptors with decentralized launch concepts, enabling defense customers to extend long-range reach in contested environments.
CesiumAstro	Developer of space communication systems supporting fast beam hopping and mission-specific payload configurations for diverse commercial and defense applications across satellite communications.
Quantic Now	Manufacturer of high-energy-density capacitors for defense and aerospace applications requiring high reliability and SWaP savings, used in radar, laser, electronic warfare, and downhole drilling systems.
Hadrian	Developer of an autonomous manufacturing platform integrating software, robotics, and full-stack capabilities to accelerate and reduce costs of aerospace and defense component production for mission-critical programs.
HawkEye 360	SIGINT and defense technology company, provides end-to-end RF signal intelligence solutions, including satellite design, data collection, and analysis, supporting U.S. and allied national security operations.
Kymeta	Developer of lightweight, low-power flat-panel satellite antennas requiring no mechanical steering, enabling mobile satellite-cellular connectivity for government, military, maritime, and public safety industries globally.
Forterra	Developer of AI-powered robotic vehicle autonomy technology for defense, commercial, and municipal use, enabling automated convoy operations, manned-unmanned teaming, and off-road navigation in mission-critical environments.
Hermeus	Developer of hypersonic aircraft using hardware-rich iterative development to radically reduce long-haul flight times, serving both commercial aviation and defense customers requiring asymmetric speed advantages.
Ursa Major	Manufacturer of liquid oxygen and kerosene propulsion engines for microsatellite launch and hypersonic applications, delivering higher performance and lower cost boost propulsion to aerospace clients.
Epirus	Developer of software-defined high-power microwave directed energy systems for counter-UAS and asymmetric threat neutralization, enabling defense organizations to defeat drone swarms and individual threats efficiently.
Impulse Space	Manufacturer of orbital maneuvering vehicles enabling multi-payload delivery to orbit from a single launch, providing companies with cost-controlled, last-mile space payload delivery and operational flexibility.
Apex	Developer of configurable, standardized satellite buses built on scalable manufacturing and mission-ready integration, enabling commercial and government customers to reduce time to orbit and expand payload capabilities.
Blue Origin	Developer of reusable rocket-powered vertical take-off and landing space vehicles, enabling commercial and government customers to expand human presence in space and access off-Earth energy and material resources.
Vast Space	Developer of artificial gravity commercial space stations with crew quarters, habitable volume, and domed windows, enabling astronauts and government agencies to live and work in space without zero-gravity health effects.
Onebrief	Developer of a visual military strategy platform offering structured, cause-and-effect plans with auto-layouts and quantitative comparison tools, enabling rapid decision-making and collaborative planning for defense clients.
Arch Global Precision	Contract manufacturer of precision-machined surgical instruments, aerospace components, and cutting tools for medical, aerospace, and defense sectors, offering prototype development, engineering, and finishing services.
Castelion	Manufacturer of hypersonic weapons systems using vertically integrated development and manufacturability-optimized designs to enable defense organizations to deploy high-performance strike systems rapidly and affordably.
Synergy Aerospace	Engineering and technical services firm offering electronic warfare, unmanned air systems, tactical EW aircraft, and DoDAF architecture and acquisition services to defense customers.
Tyto Athene	Provider of mission-critical communications systems integration services for defense, intelligence, and civilian agencies, specializing in networking, security, and digital transformation to enhance national readiness.
K2 Space	Developer of high-power satellite buses providing propulsion and power for defense, commercial, and scientific payloads across LEO, MEO, GEO, and cislunar orbits with speed and flexibility.
UVeye	Developer of AI-driven vehicle inspection systems using high-resolution cameras and computer vision to detect concealed weapons and contraband, serving government agencies, automakers, rental companies, and logistics centers.
Terran Orbital	Manufacturer of end-to-end satellite solutions combining design, production, launch planning, and on-orbit support for military, civil, and commercial customers, operating primarily in the United States.
Capella Space	Developer of SAR satellite imagery technology providing continuous global coverage through clouds, smoke, and darkness, enabling defense, disaster relief, and maritime operations to conduct infrastructure monitoring and geospatial analysis.

Source: Pitchbook

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Completed 08 Jun 2026 01:41 PM EDT

Disseminated 08 Jun 2026 02:48 PM EDT